Introduction to Biosensors

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Tópicos em Biofísica e Bionanossistemas
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Mestrado em Engenharia Física

Outline

- What are biosensors?
- Biosensors with electrical transduction
  - Field-effect transistors
  - electrochemical sensors
- Solid-liquid interfaces: the electrical double layer
- Techniques to study the solid-liquid interface:
  - The quartz crystal microbalance
  - The Langmuir isotherm
  - Electrokinetic phenomena and the Zeta potential measurement
  - DLVO theory
- 2D materials and interfaces
- Biosensor development and fabrication

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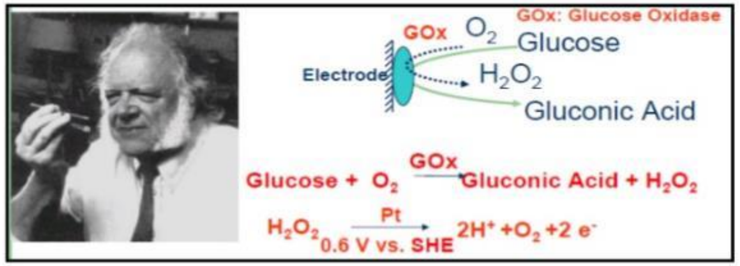
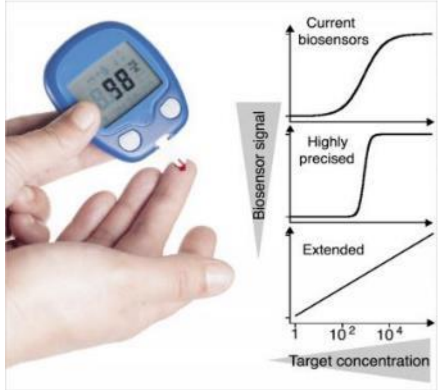
Biosensors

Professor Leland C. Clark (1918 – 2005)



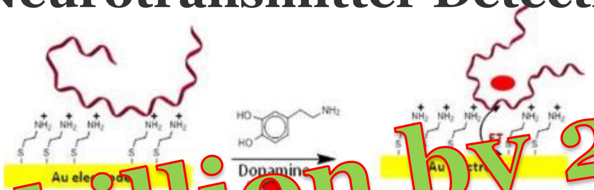
First description of a biosensor in 1962 - an amperometric enzyme electrode for glucose.

Glucose



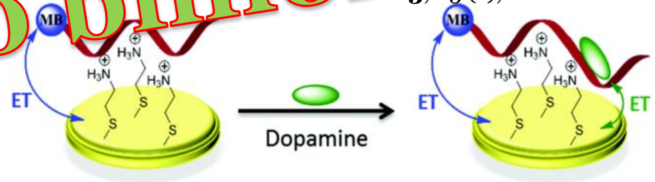
The Plaxco Group At University Of California Santa Barbara

Neurotransmitter Detection



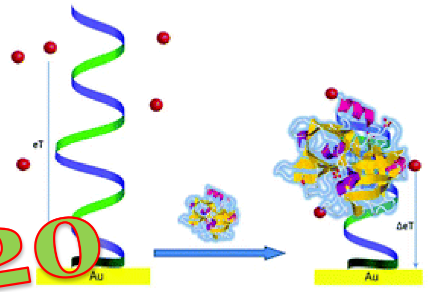
E. Farjami et al. Anal. Chem. 2013, 85 (1), 121–128

$20 billion by 2020



I. Martos, R. Campos and Elena E. Ferapontova, Analyst, **2015**, 140, 4089-4096

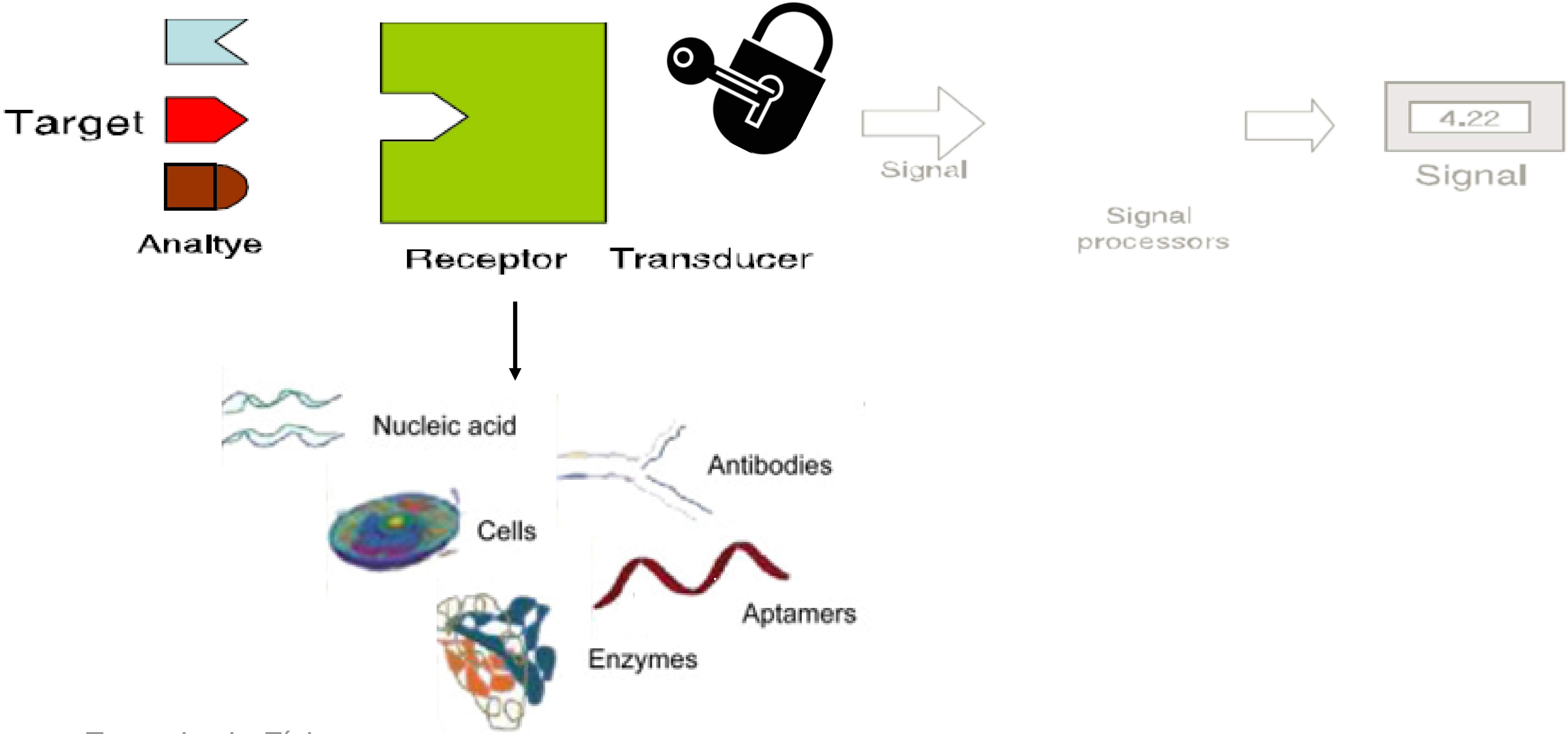
Cancer Biomarker



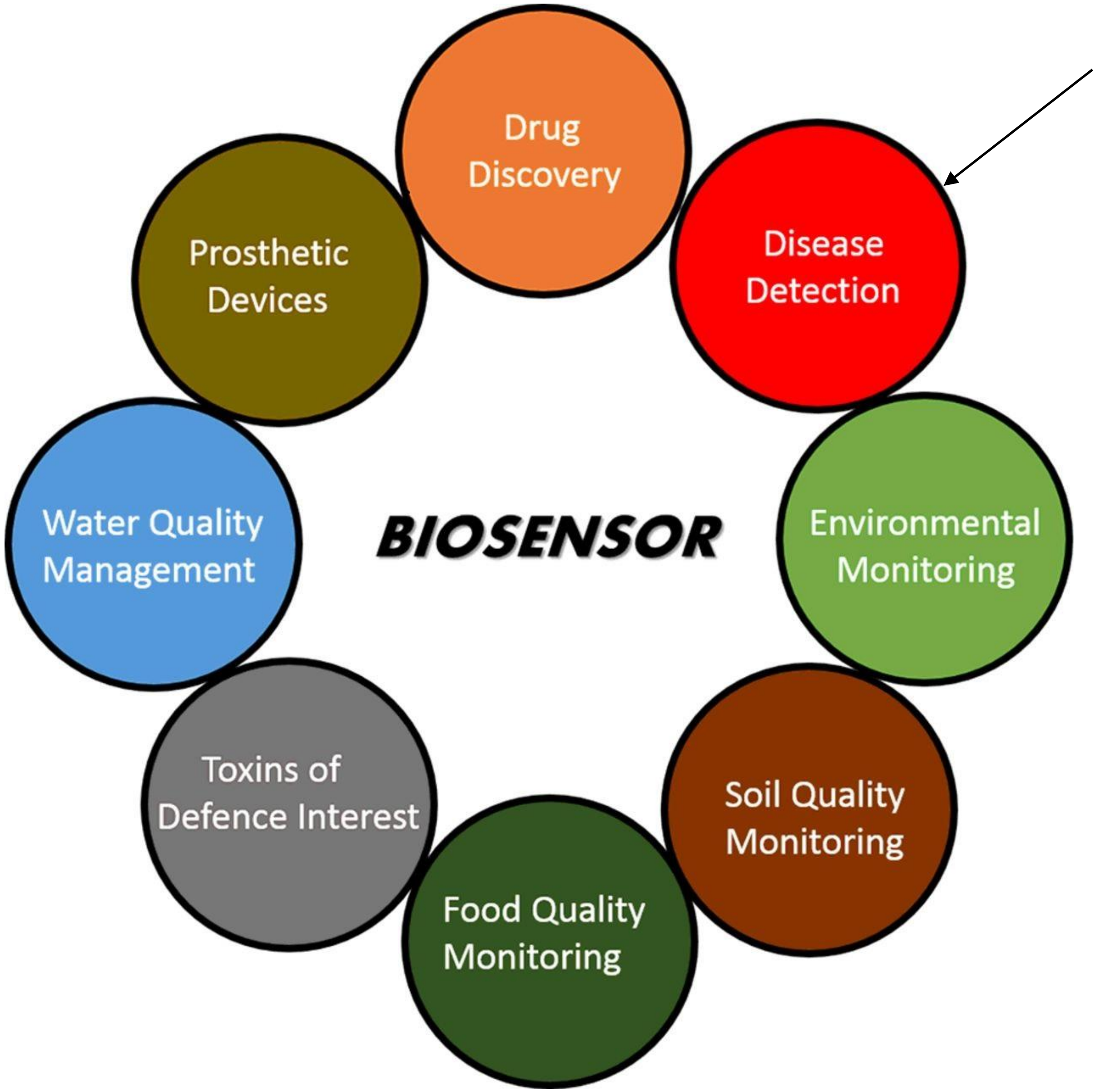
M. Jarczewska *et al.*, Analyst, **2015**, 140, 3794-3802

What are biosensors?

- **Biosensor** = biological element + **sensor**



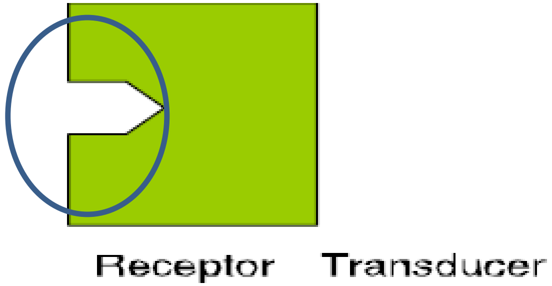
Biosensors’ versatility

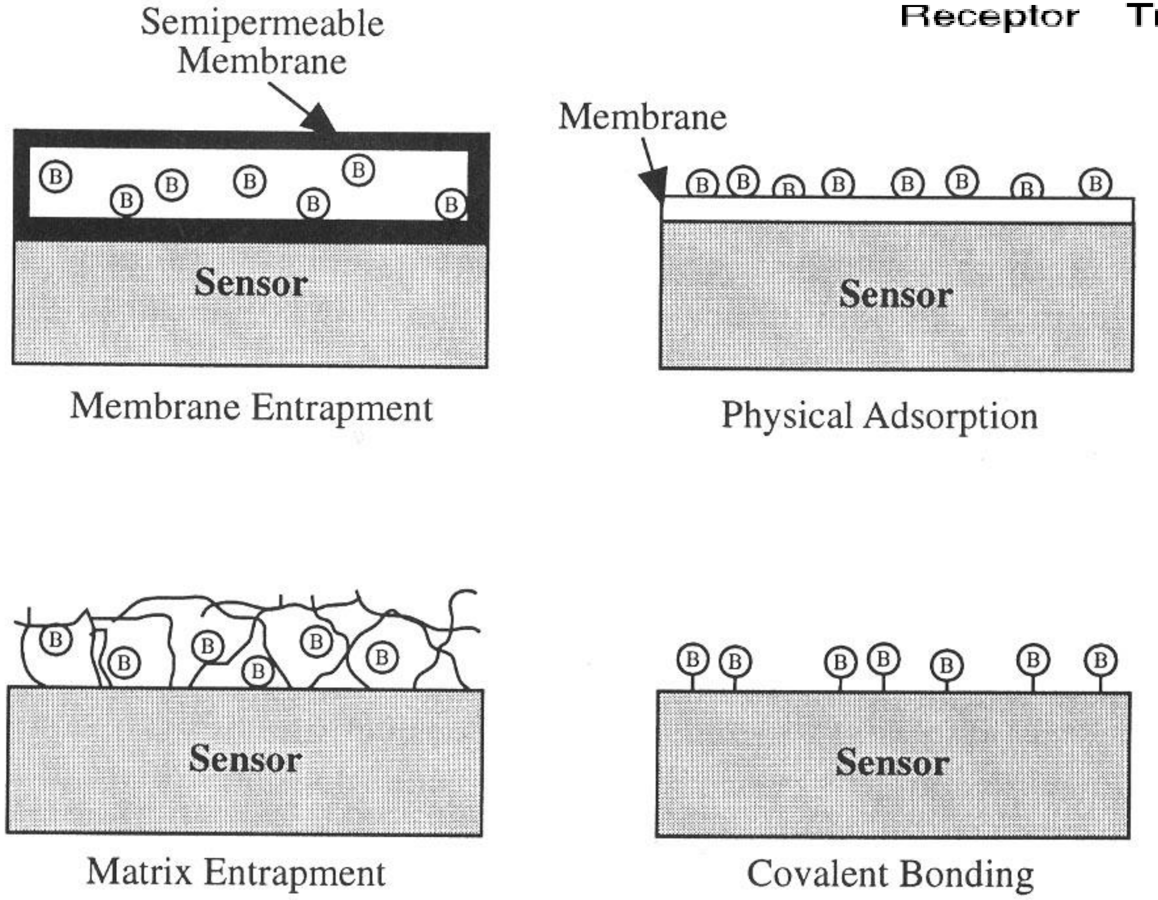


https://www.elprocus.com/what-is-a-biosensor-types-of-biosensors-and-applications/

Functionalization

- Immobilization of Biological Component





Biosensors

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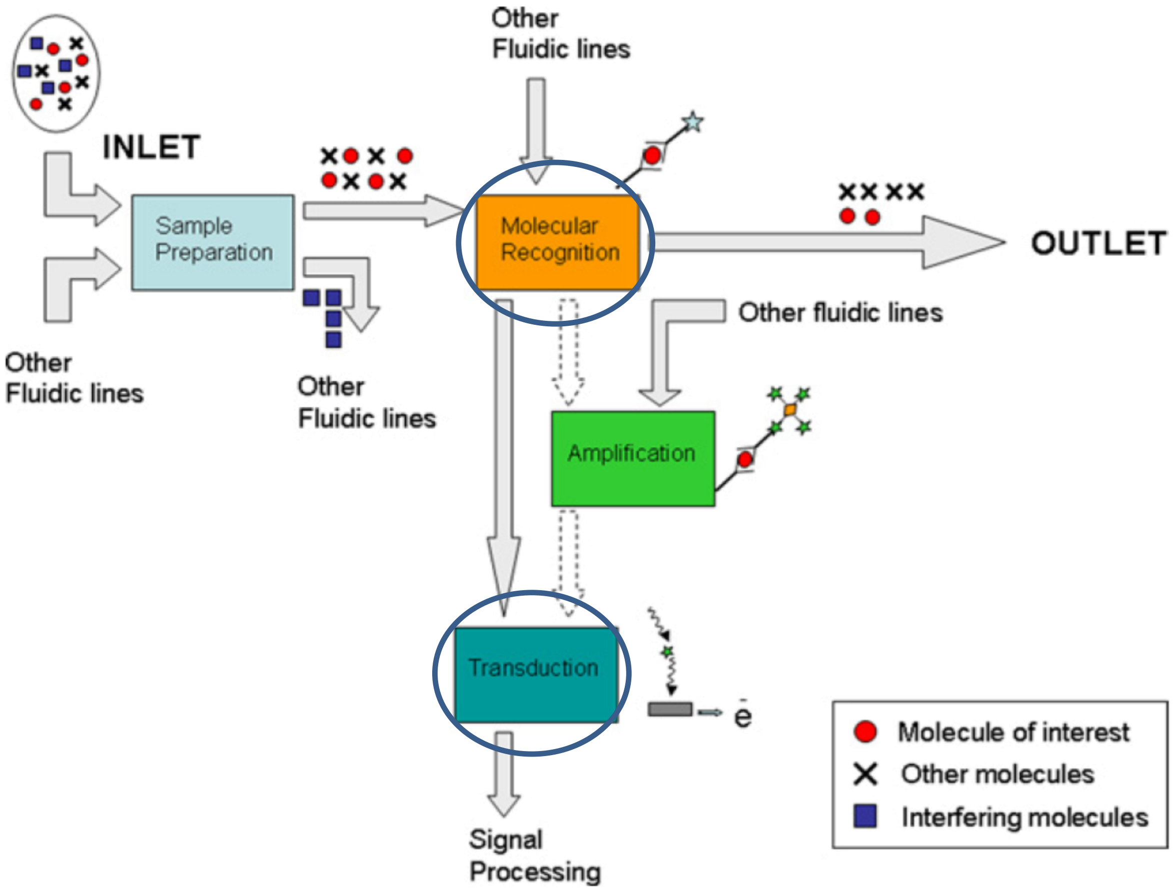
Lab-on-chip systems for integrated bioanalysis



J.P. Conde et al., *Essays in Biochemistry* (2016) **60** 121–131

Transducers can use:

- Optical detection
- Electronic detection
- Thermal detection
- Piezoelectric detection

Detection can be:

- Label-free
- Labeled

Types of Biosensors

- *based on analytes and detection mode*

Biorecognition Element	Biosensor
Enzymes	Enzyme electrode
Proteins	Immunosensor
Antibodies	
DNA	DNA sensor
Microbial cells	

Transducer	Measured property
Electrochemical	Potentiometric, Amperometric, Voltametric
Electrical	Surface conductivity, electrolyte conductivity
Optical	Fluorescence
Mass sensitive	Frequency
Thermal	Heat of reaction

Biosensor Technology and Medicine

- Diagnosis of disease

Medical Diagnostic Testing

↓

Clinical information

↓

Clinical decision

The 5 typical characteristics of biosensors:

1. Selectivity
2. Sensitivity
3. Linearity
4. Stability
5. Reproducibility

Features of a biosensor

- static and dynamic attributes

e.g. Light sensitive sensor

What is **sensitivity** and why are **sensitivity** statements often misleading?

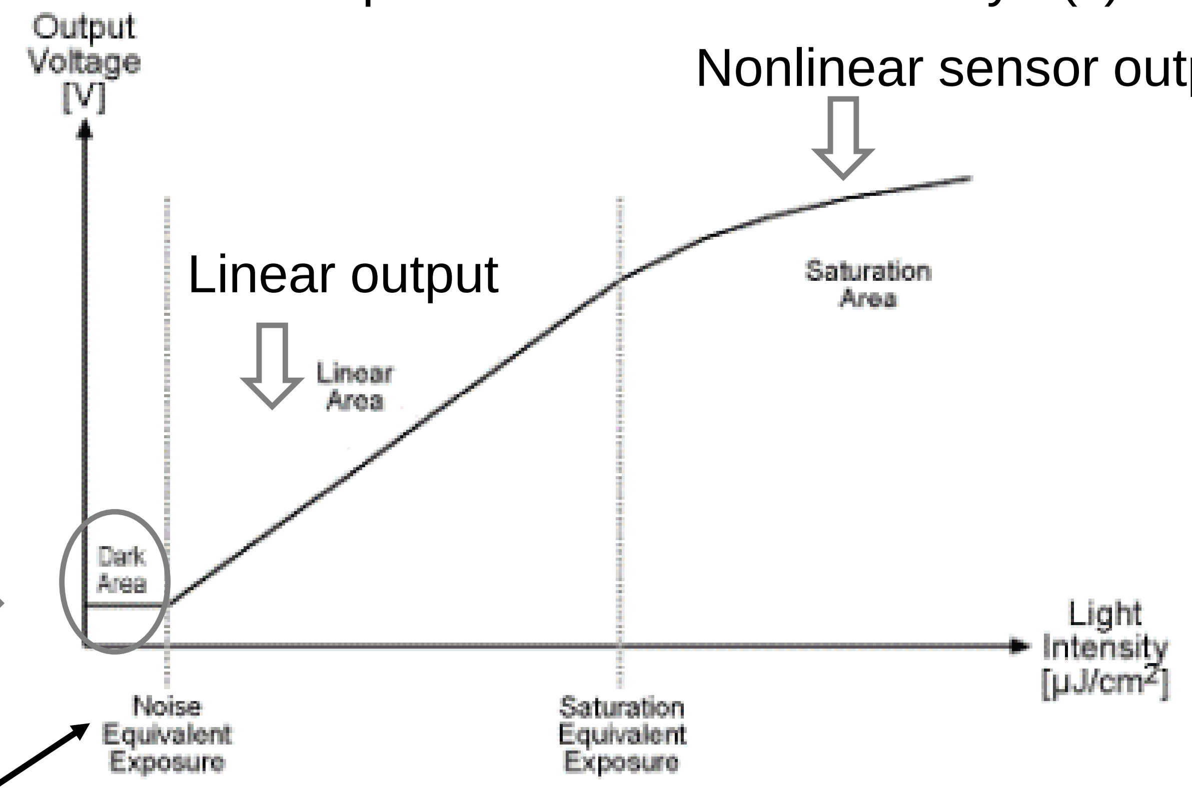
⇩

The slope of the calibration curve y=f(x)

RESPONSE CURVE

Nonlinear sensor output

Linear output

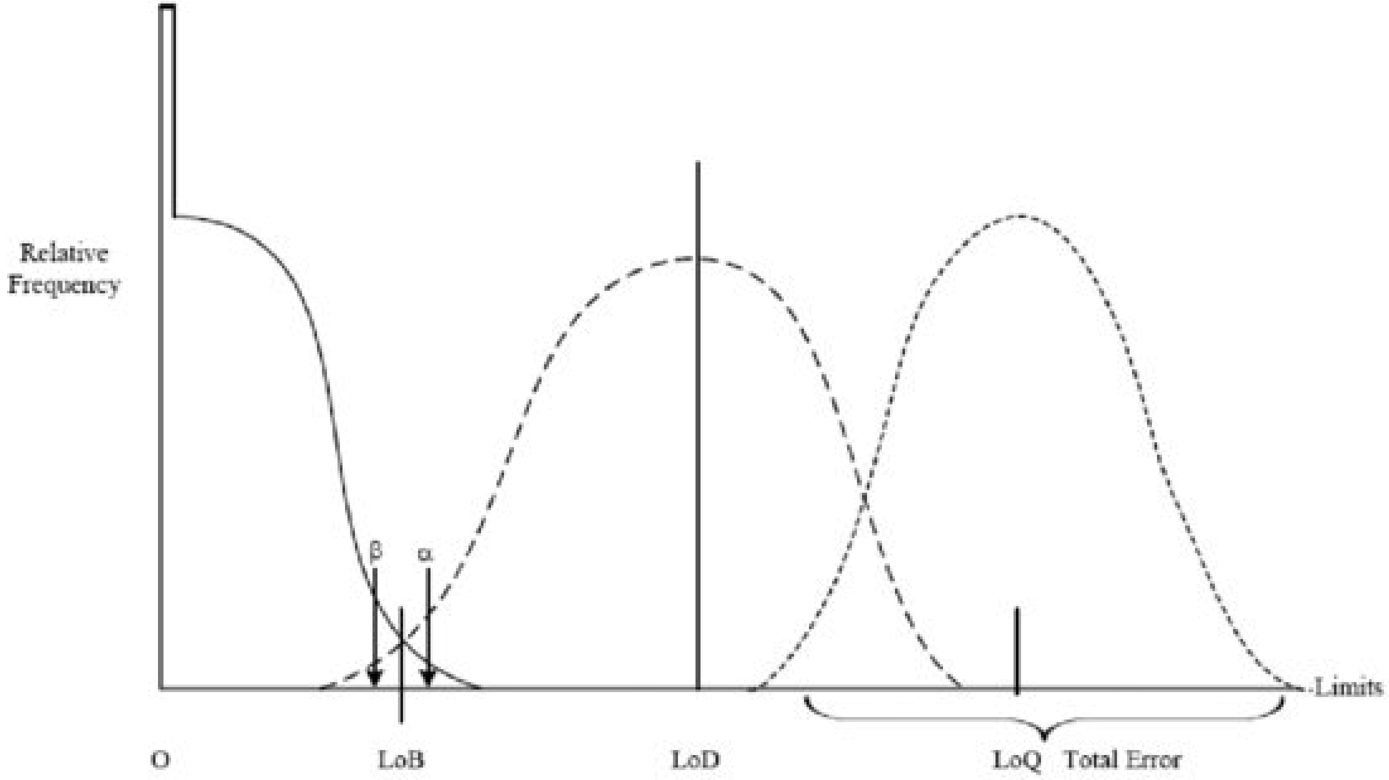


Low sensor output
(noisy and unpredictable)

Output of the sensor begins to increase *(predictably)*

Features of a biosensor

- Analytical performance: LOB, LOD, LOQ



- **Limit-of-Detection (LoD)** corresponds to the lowest analyte concentration that can be reliably distinguished from the **Limit-of-Blank (LoB),** which is determined from replicate measurements of blank samples
- The **Limit-of-Quantification (LOQ)** is the lowest analyte concentration that can be quantitatively detected with a stated accuracy and precision

Features of a biosensor

- Analytical performance: LOB, LOD, LOQ

Examples

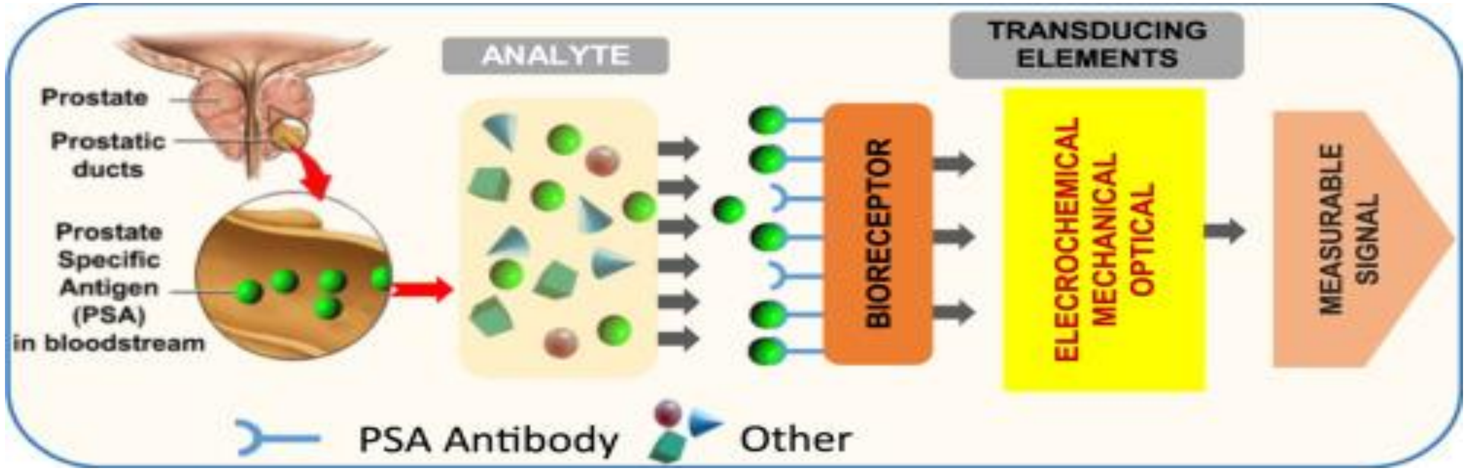
Reminder: The minimum amount of analyte that can be detected by a biosensor defines its limit of detection (LOD) or sensitivity.

⇩

In a number of medical monitoring applications, a biosensor is required to detect analyte concentration of as low as **ng/ml** or even **fg/ml** to confirm the presence of traces of analytes in a sample.

⇩

For instance, a prostate-specific antigen (PSA) concentration of 4 ng/ml in blood is associated with prostate cancer for which doctors suggest biopsy tests.

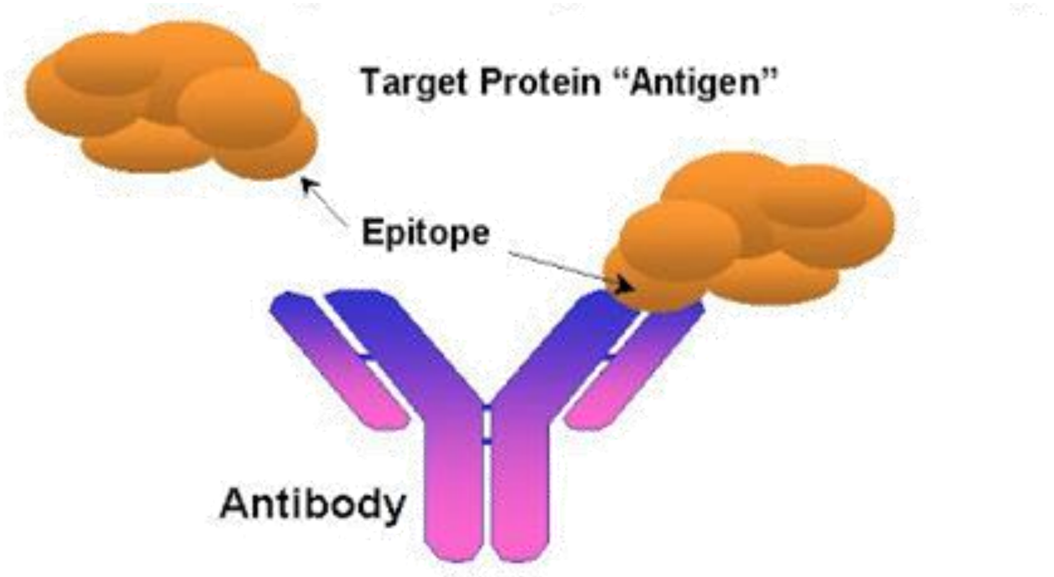


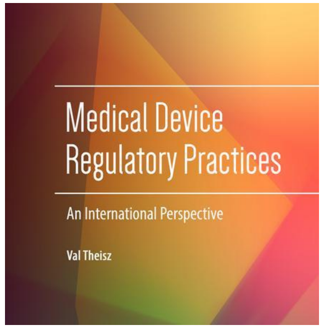
Features of a biosensor

static and dynamic attributes

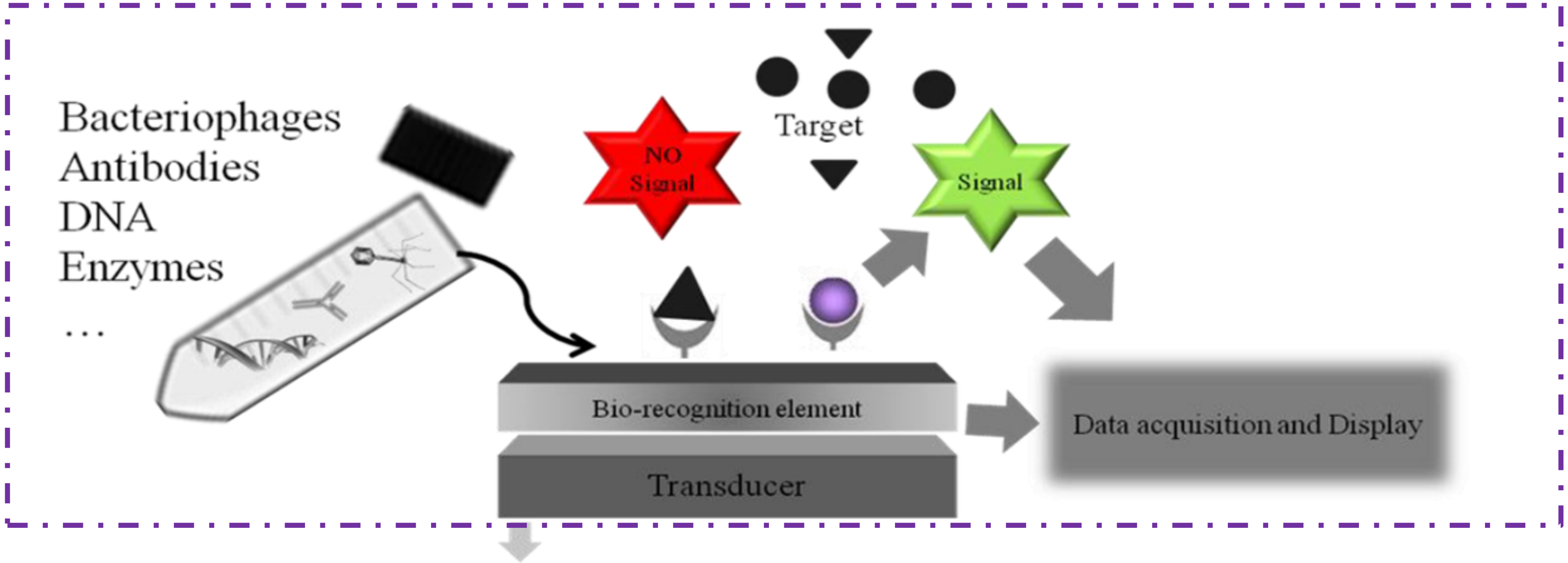
Selectivity: The potential of a bioreceptor to detect the specific analyte (target) in a mixture of sample and contaminants.

e.g. Antibody-antigen interaction





"From a medical device regulatory perspective, it is critical to define **key quality system requirements** and ensure that they are meet or exceeded enabling the development of a **safe and effective diagnostics device**".



Optical (e.g. SPR, Fiber Optic)

Mass-based (e.g. Piezoelectric, Magnetoelastic)

Electrochemical (e.g. Amperometric, Potenciometric, Impedimetric)

Magnetoresistives

Diagnostic biosensors in medicine

- Decrease detection time
- Lower the detection limit
- Increase specificity and sensitivity
- High-throughput analysis of biologically important molecules

= Powerful tools for in vitro diagnostics (IVDs)

Risk of misdiagnosis:

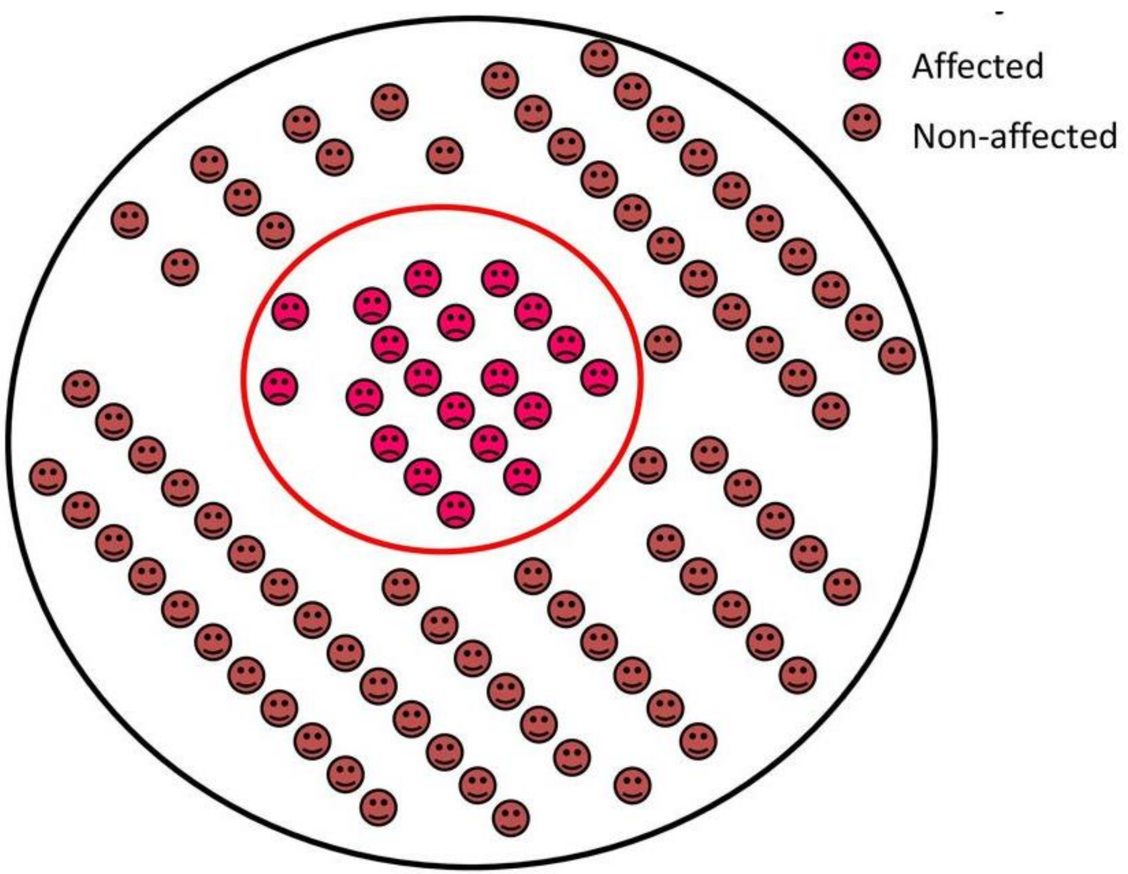
False-positive or False-negative

International Standardization Organization (ISO) and Regulatory Agencies cover main aspects of Biosensors manufacturing, validation and use.

Diagnostic Test

- The perfect diagnostic test

Tests should be 100% accurate but this is an unrealistic scenario



Analysis of Diagnostic Tests

Used by physicians to help diagnose an illness, injury, disease or any other type of medical condition.

Test result *(four possible outcomes)*

Means: Disease

True Condition	Positive	Negative
Positive	True Positive (A)	False Negative (C)
Negative	False Positive (B)	True Negative (D)

Means: Healthy

False-positive: Healthy people incorrectly identified as sick

False-negative: Sick people incorrectly identified as healthy

Analysis of Diagnostic Tests

- Diagnostic test accuracy

The most common measures of **diagnostic test accuracy** are <u>sensitivity</u> (true positive rate) and <u>specificity</u> (true negative rate).

= test effectiveness

What is Sensitivity ?

$$Sensitivity = \frac{Number\ of\ True\ positive\ test}{(Number\ of\ True\ positive + Number\ of\ False\ negative)}$$

OR

$$Sensitivity = \frac{Number\ of\ True\ Positive\ Test}{Total\ number\ of\ individuals\ with\ the\ disease\ in\ a\ population}$$

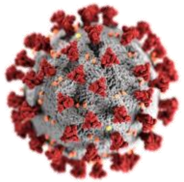
What is Specificity ?

$$Specificity = \frac{Number\ of\ True\ Negative\ Test}{Number\ of\ True\ negative + Number\ of\ false\ positive}$$

OR

$$Specificity = \frac{Number\ of\ True\ Negative\ tests}{Total\ number\ of\ healthy\ individuals\ in\ a\ population}$$

Analysis of Diagnostic Tests



- Example

Assumption: You have a new rapid diagnostic test being evaluated for the screening of COVID-19, on the specific antibodies produced against the virus, SARS-CoV-2. You have a sample size of 600 people and by validity, there are samples that you know definitely have the disease (480) and/or healthy individual samples from the disease in question (120).

	Positive	Negative	Total
True Positive	480	–	480
False Positive	15	–	15
True Negative	–	100	100
False Negative	–	5	5
Total	495	105	600

True Positive Rate (TPR)

True Negative Rate (TNR)

Sensitivity = 480/(480+5)= 0.98

- Therefore, the test has a 98% sensitivity.

Specificity = 100/(100+15)=0.87

- Therefore, the test has 87% specificity.

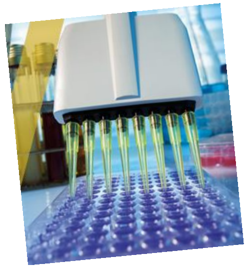
Note: A test that has a high sensitivity but low specificity causes many patients who do not have the disease being told of a possibility that they have a disease, subjecting them to further testing

Gold Standard Tests

"Gold Standard" is the method used to obtain a definitive diagnosis, and is used to define **true disease status** against which the results of a new diagnostic test are compared.

Examples

Target Disorder	**Gold Standard**
breast cancer	excisional biopsy
prostate cancer	transrectal biopsy
coronary stenosis	coronary angiography
myocardial infarction	catheterization
strep throat	throat culture

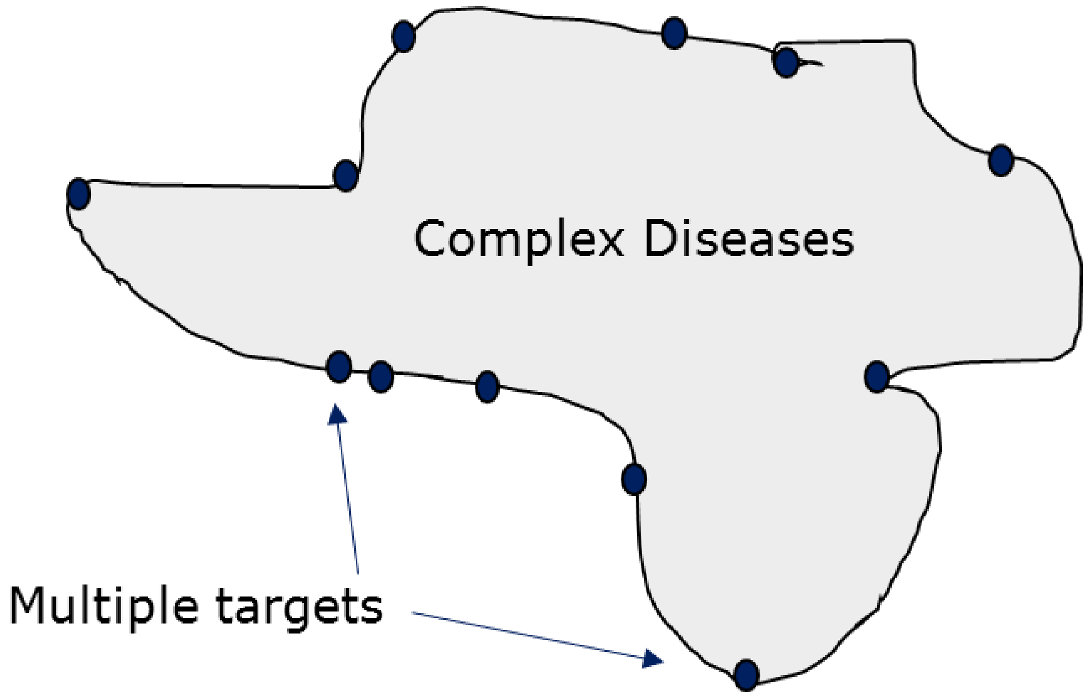


Enzyme-Linked ImmunoSorbent Assay (ELISA)

Immunological procedures in which Ag-Ab reaction is monitored by enzyme measurements

Complex diseases

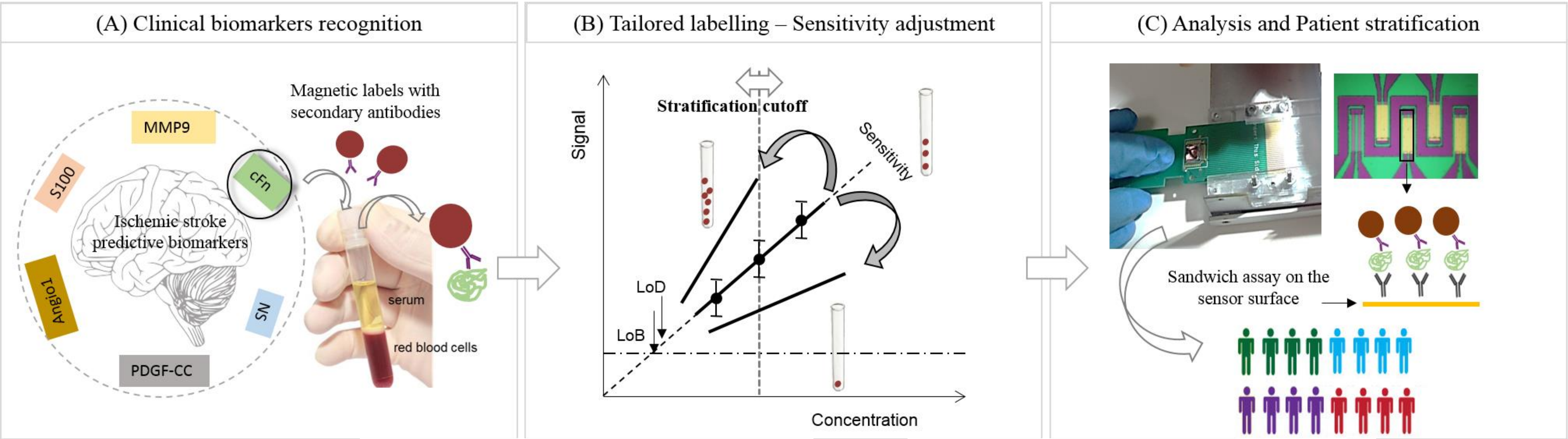
- Panel of biomarkers and clinical cutoff values

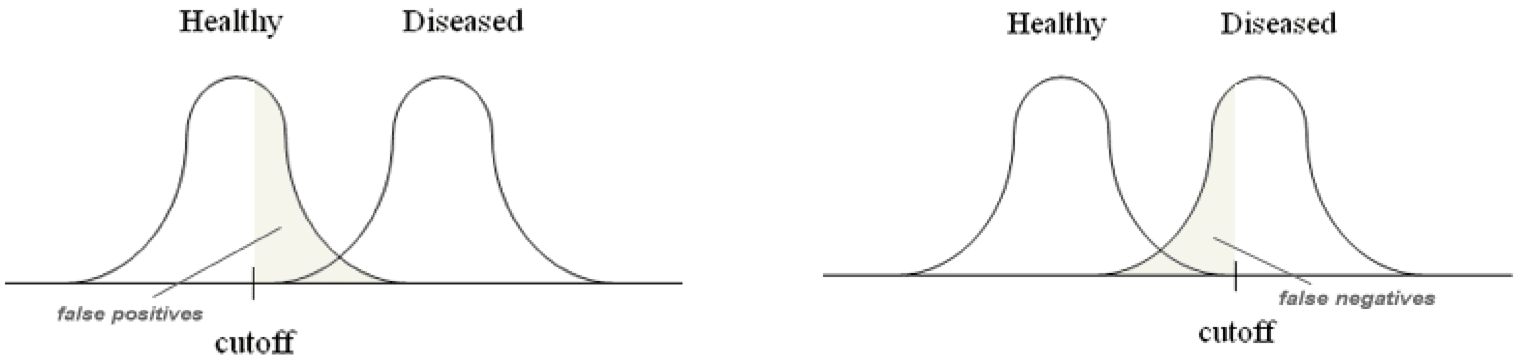


Complex diseases

- Panel of biomarkers and clinical cutoff values

is not universal and should be determined for each disease condition





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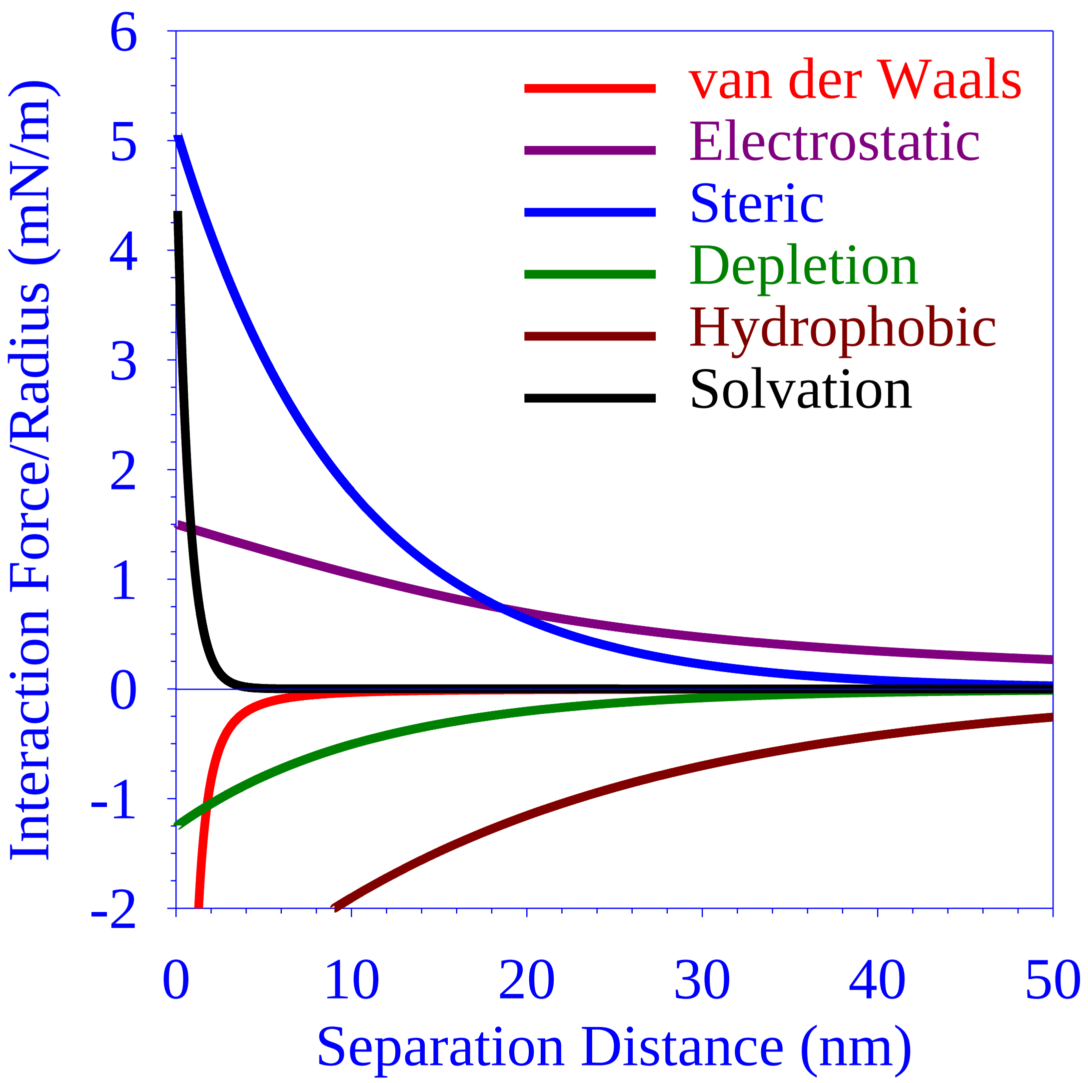
Intermolecular Forces

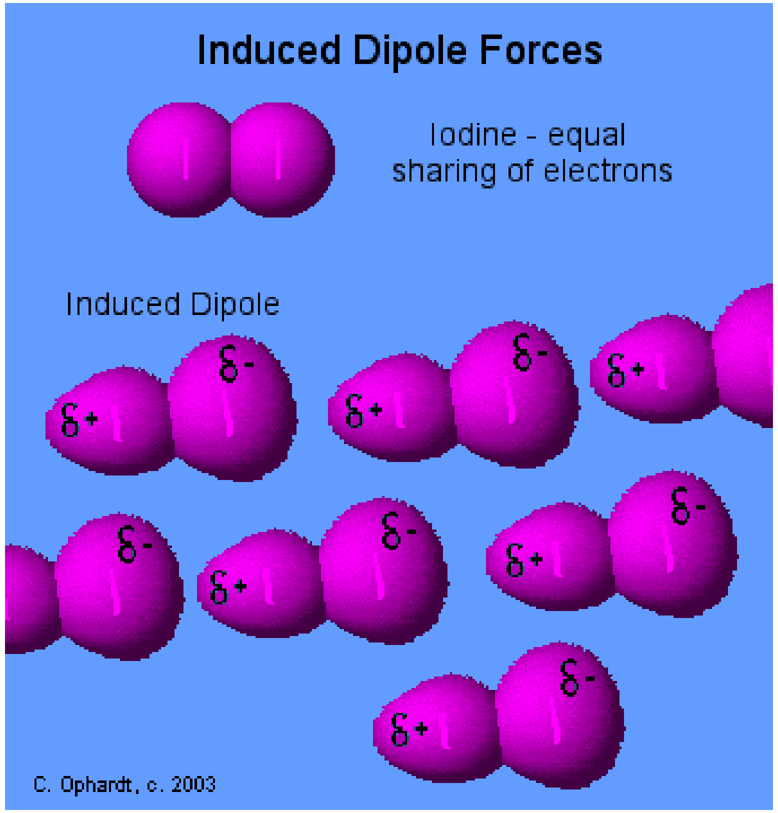


Repulsive Forces (Above X-axis)

Attractive Forces (Below X-axis)

Intermolecular Forces

Induced Dipole Forces → London Dispersion force



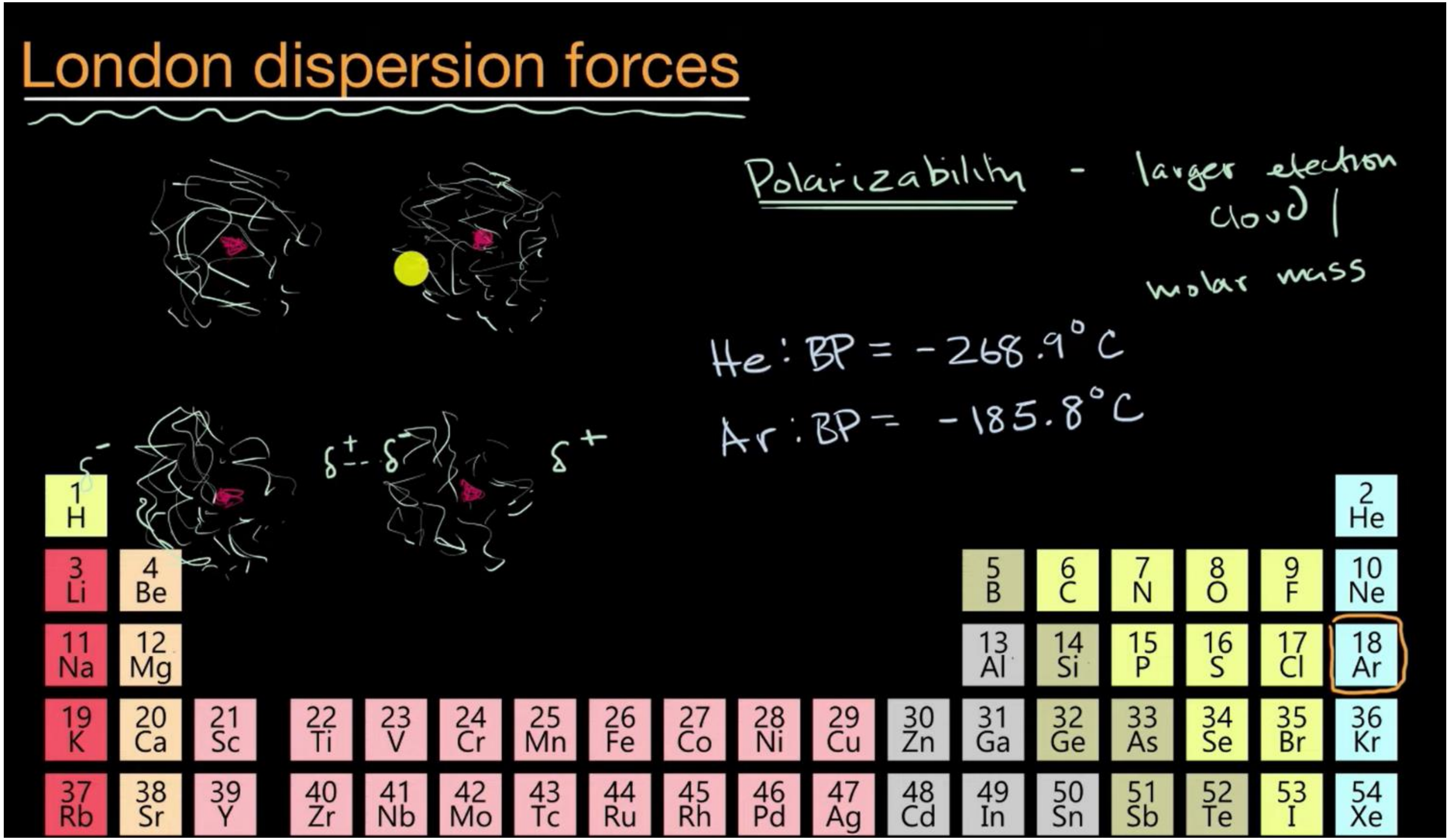
- "Temporary dipoles" are formed by the shifting of electron clouds within molecules.
- These temporary dipoles attract or repel the electron clouds of nearby non-polar molecules.

London Dispersion Force, also called induced dipole-induced dipole, is the weakest intermolecular force. It is the interaction between two nonpolar molecules. For example: interaction between methyl and methyl group: $(-CH_3 \leftrightarrow CH_3-)$. (approximate energy: 5 KJ/mol ≈ 0.050 eV/CH_3)

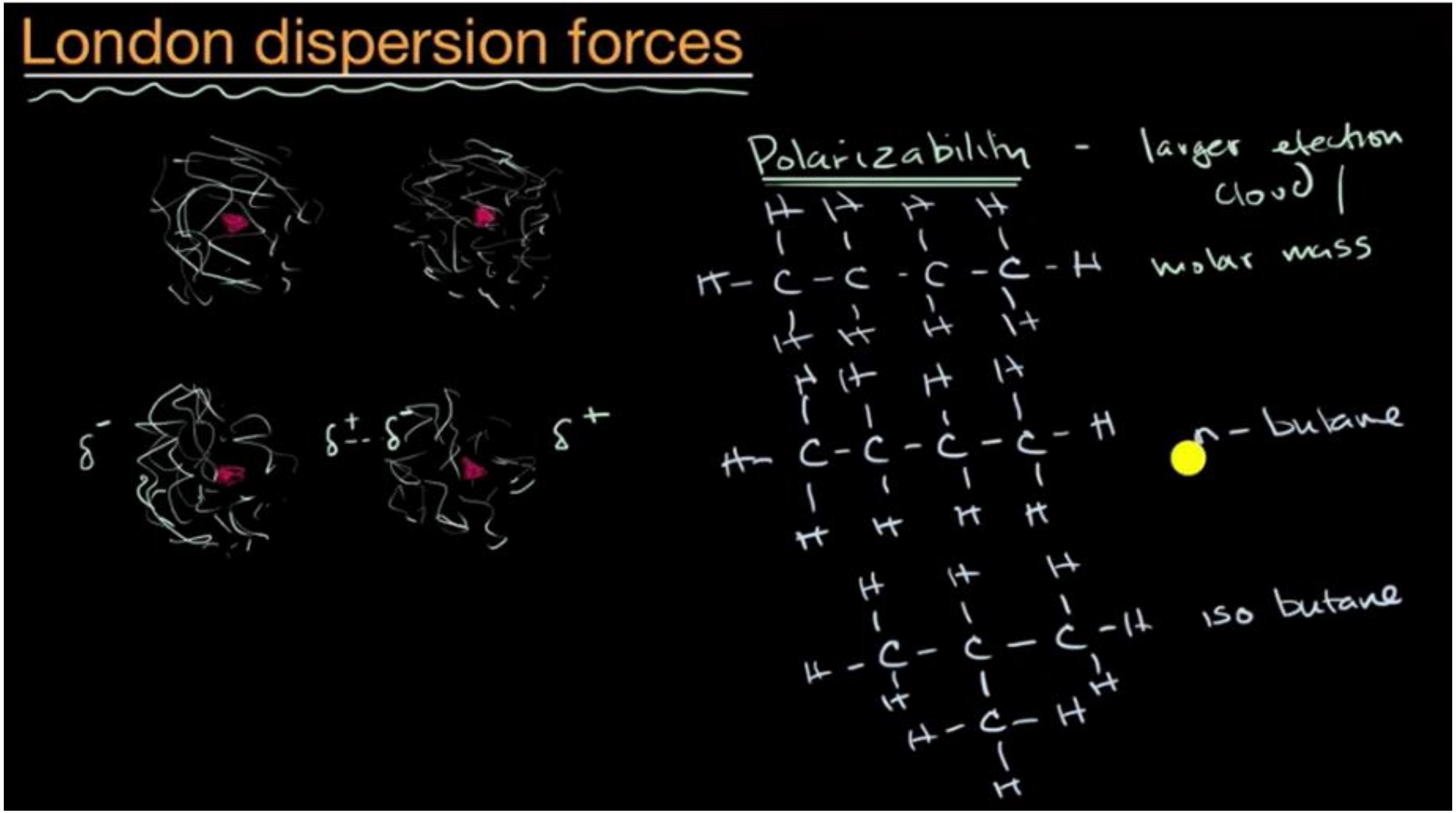
$$V = -\frac{3}{4}\frac{\alpha^2 I}{r^6}$$

- α is the polarizability of nonpolar molecule.
- r is the distance between the two molecule.
- I = the first ionization energy of the molecule.

Why do He and Ar have so different BPs?



London Dispersion Forces



Credits: Khan Academy

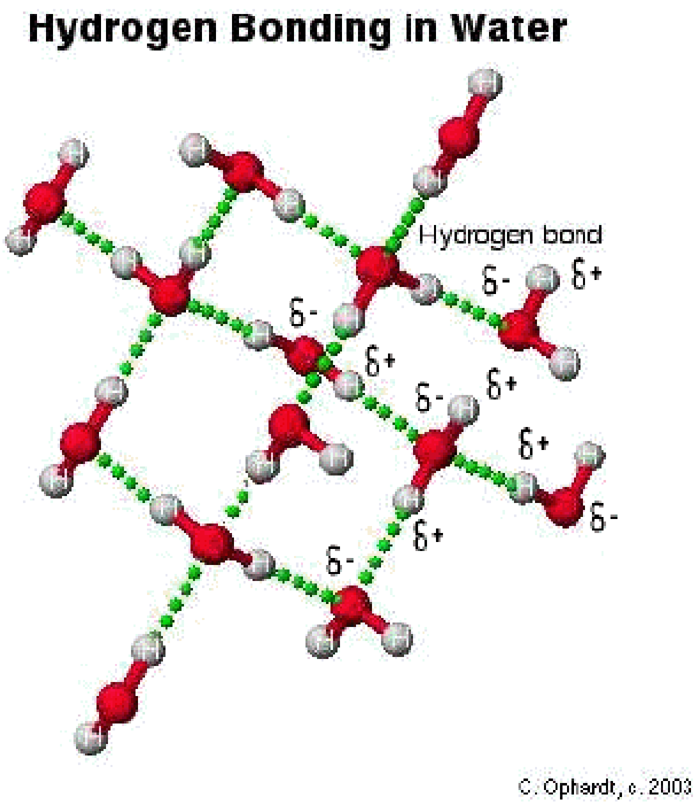
Intermolecular Forces

Dipole-Dipole Interactions

$$V = -\frac{2}{3}\frac{\mu_1^2 \mu_2^2}{(4\pi\varepsilon_0)^2 r^6}$$

μ is the permanent dipole moment of the molecule 1 and 2

Hydrogen Bonding



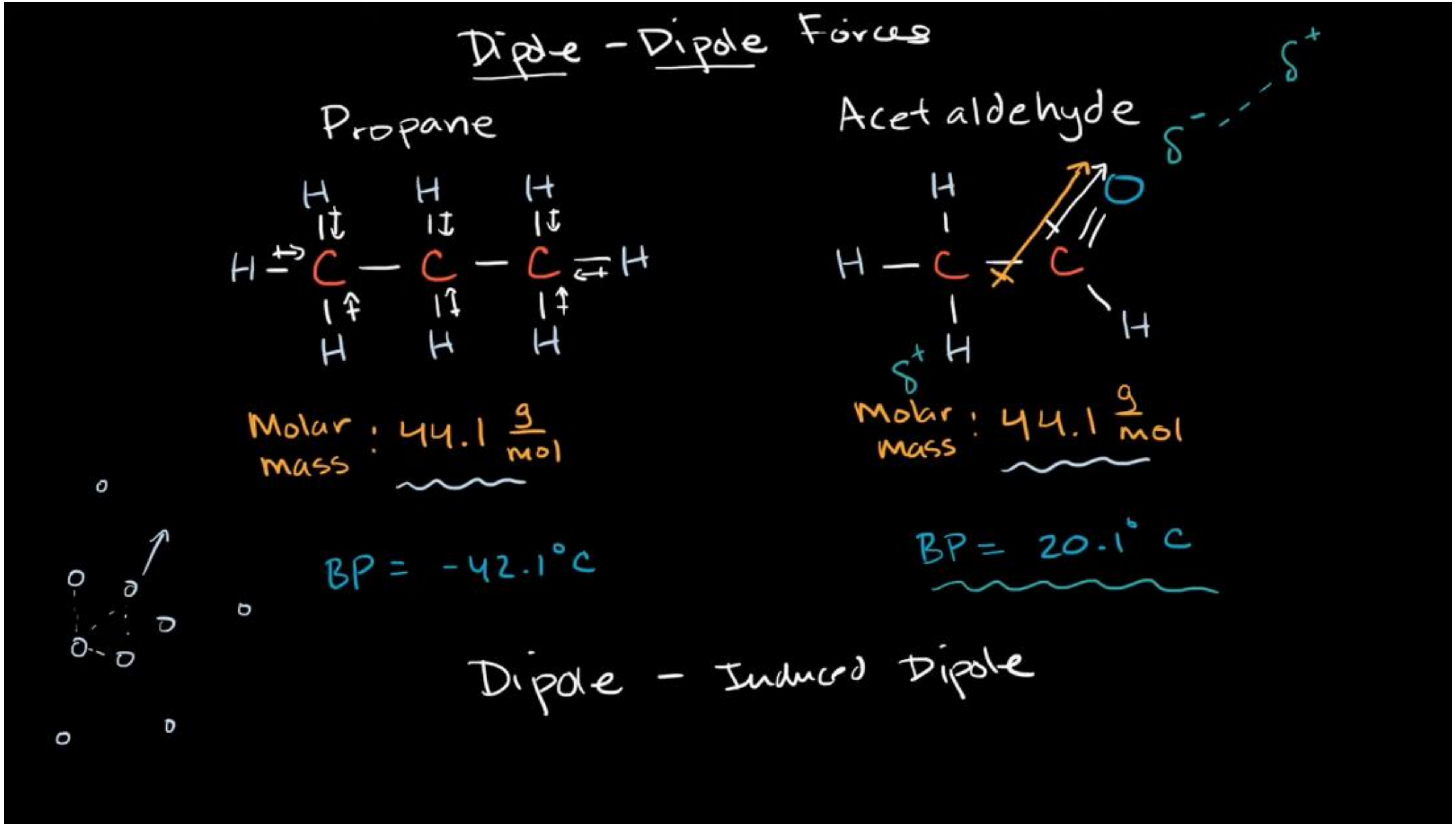
Ion-Dipole Interactions

$$Na^+ \leftrightarrow (OH_2)_n$$

$$V = -\frac{q\mu}{(4\pi\varepsilon_0) r^2}$$

- *r* is the distance of separation.
- *q* is the charge of the ion (only the magnitude of the charge is shown here.)
- μ is the permanent dipole moment of the polar molecule.

Dipole-Dipole Forces



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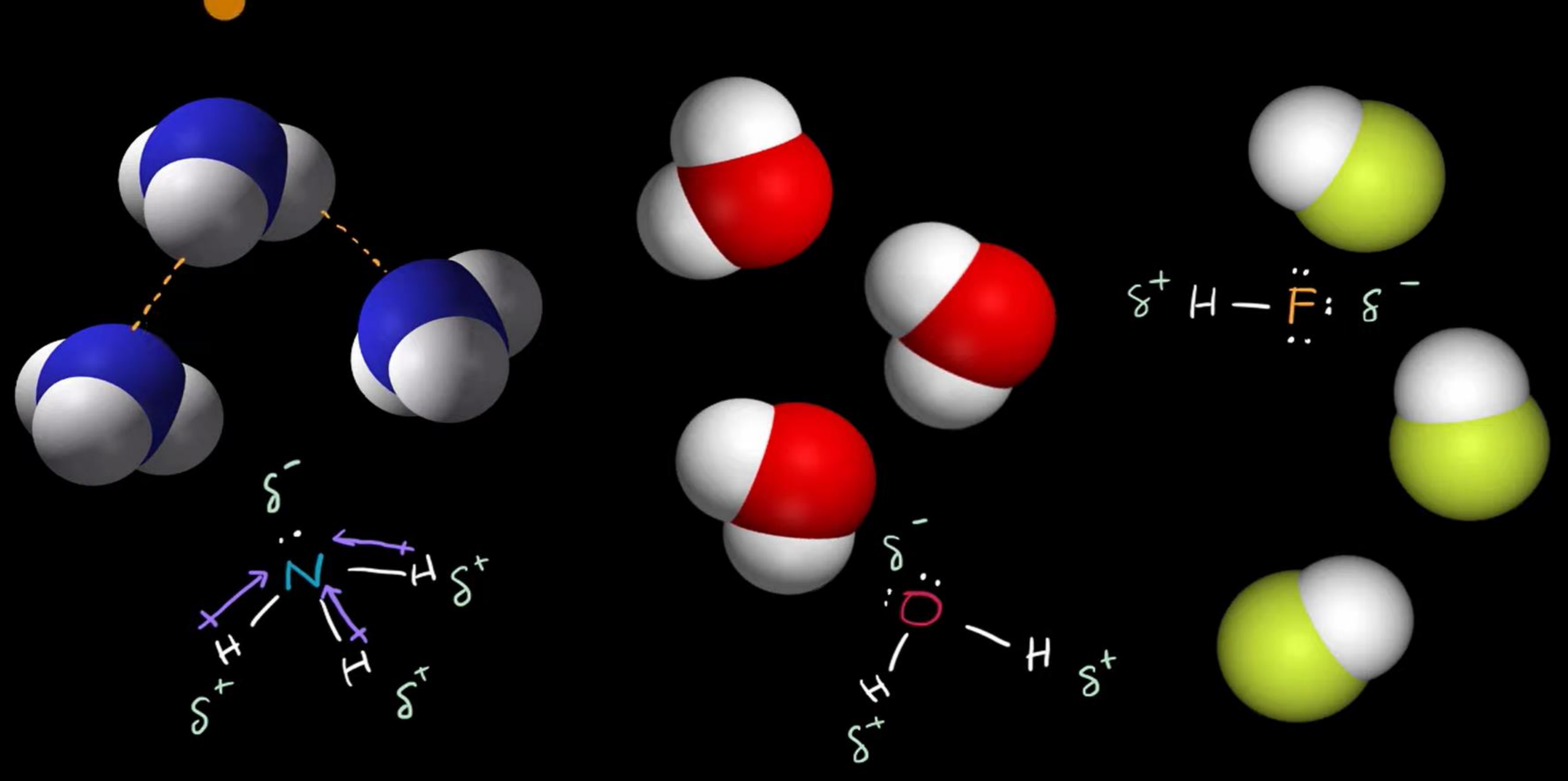
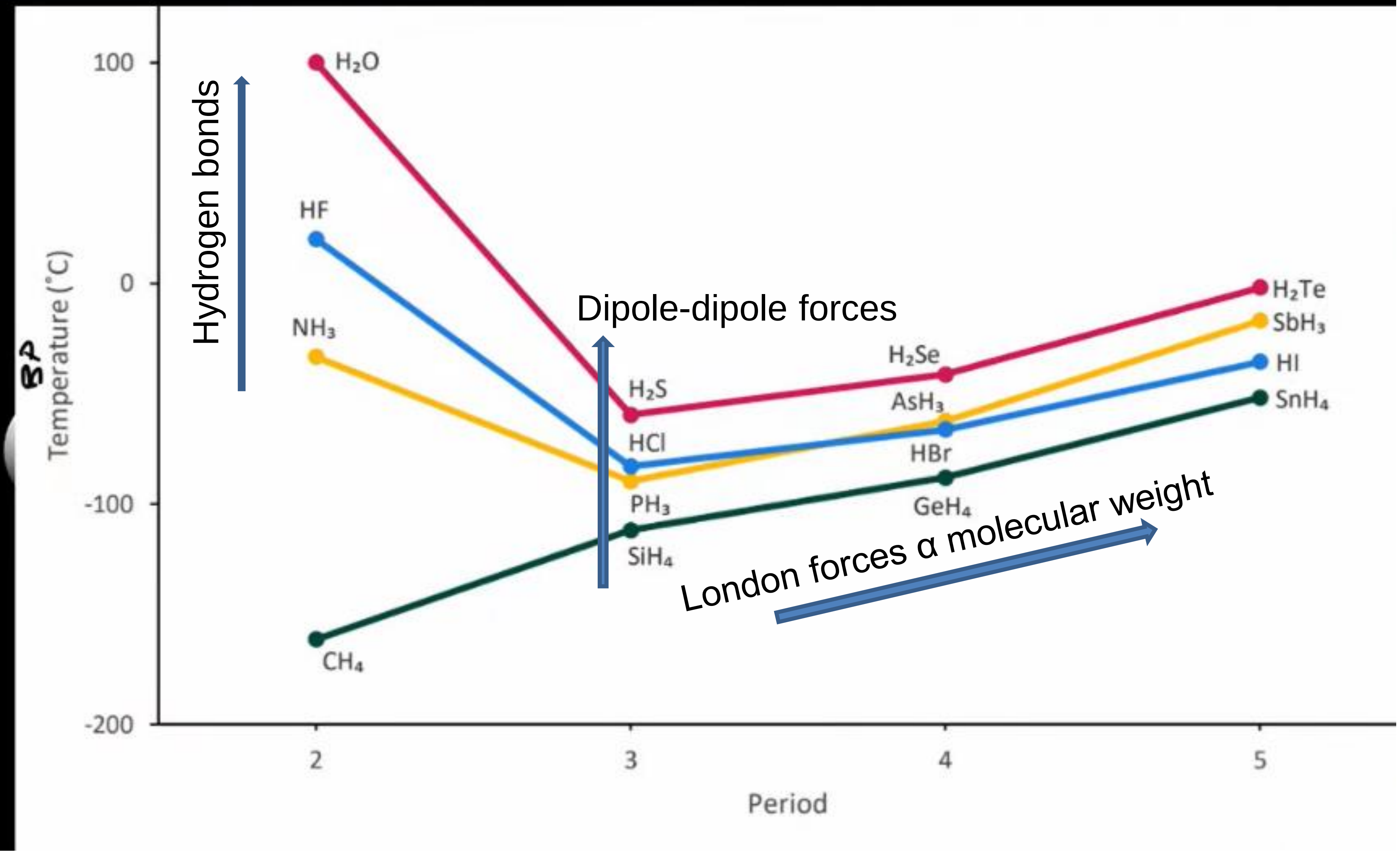
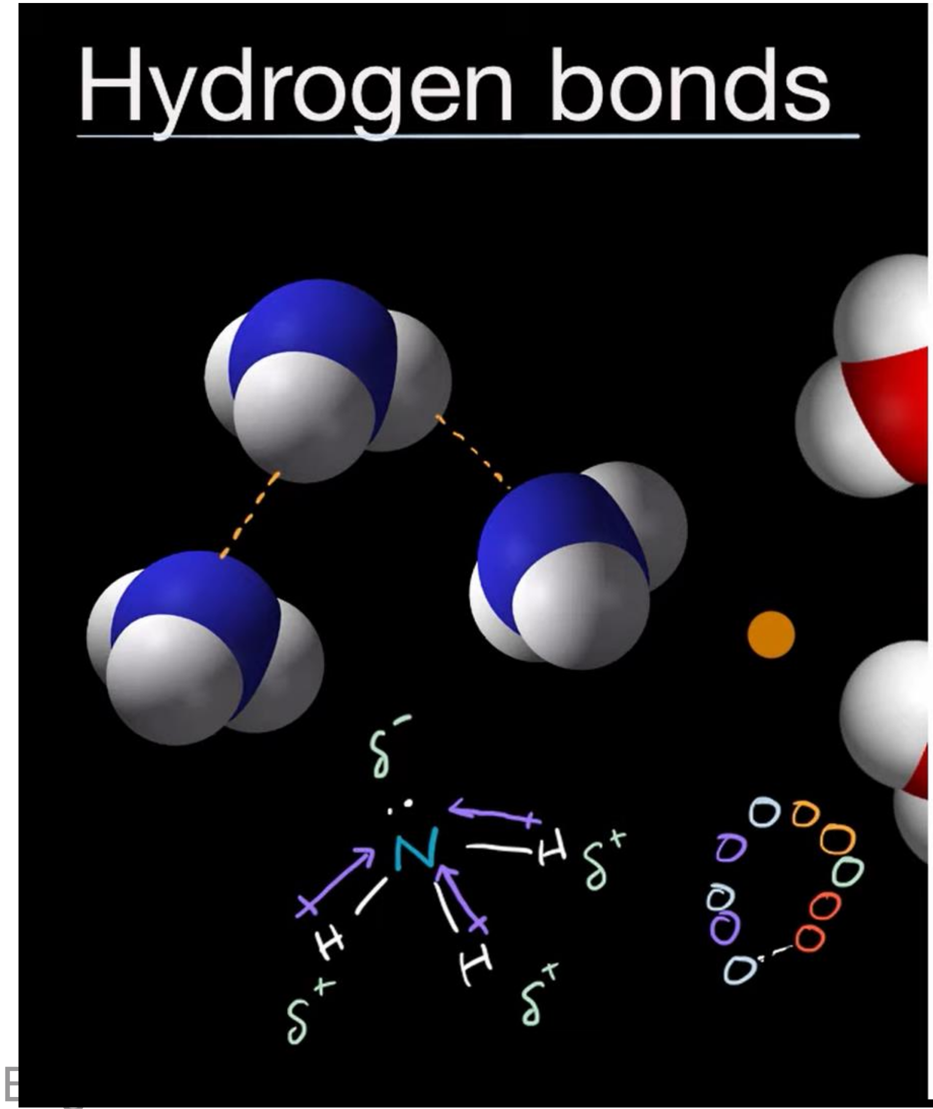
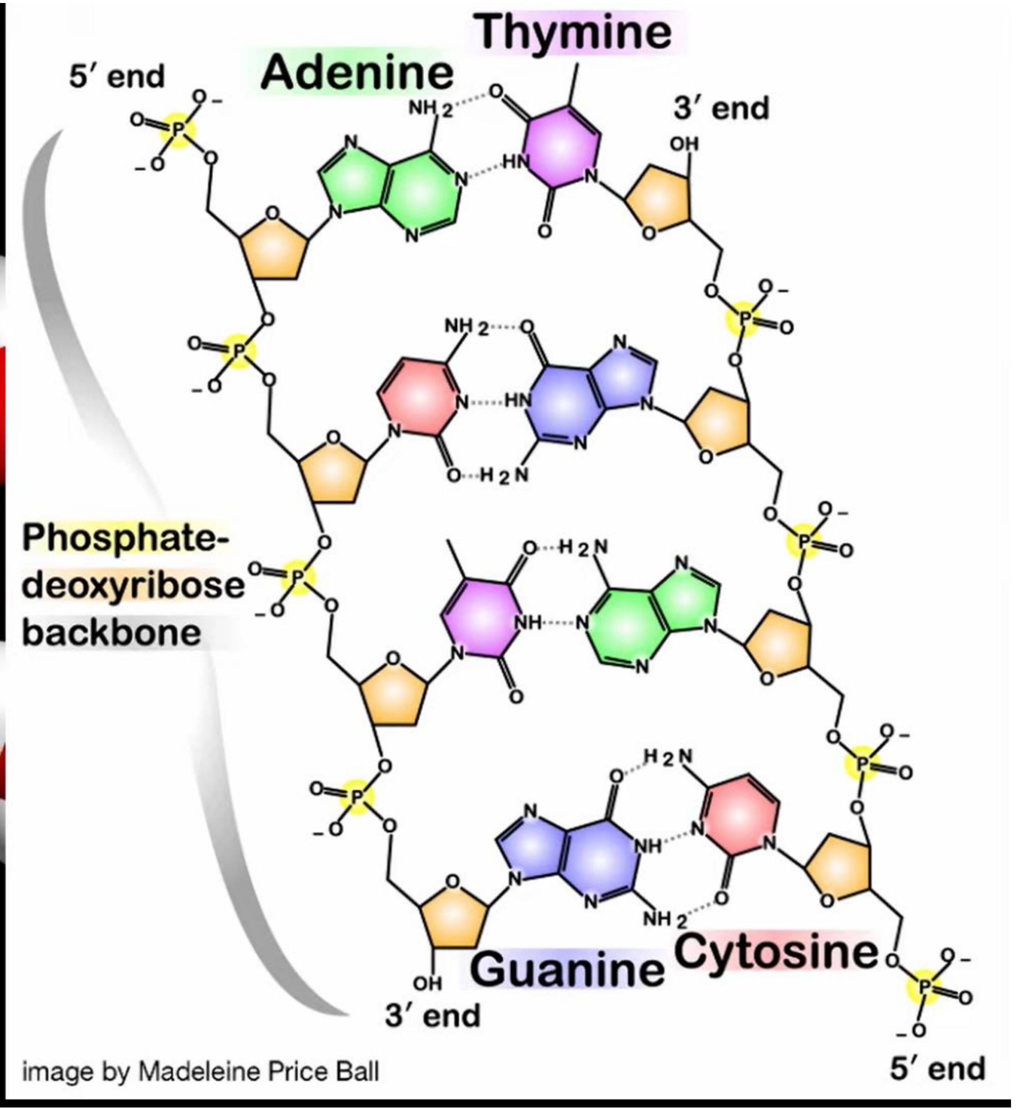
Hydrogen bonds

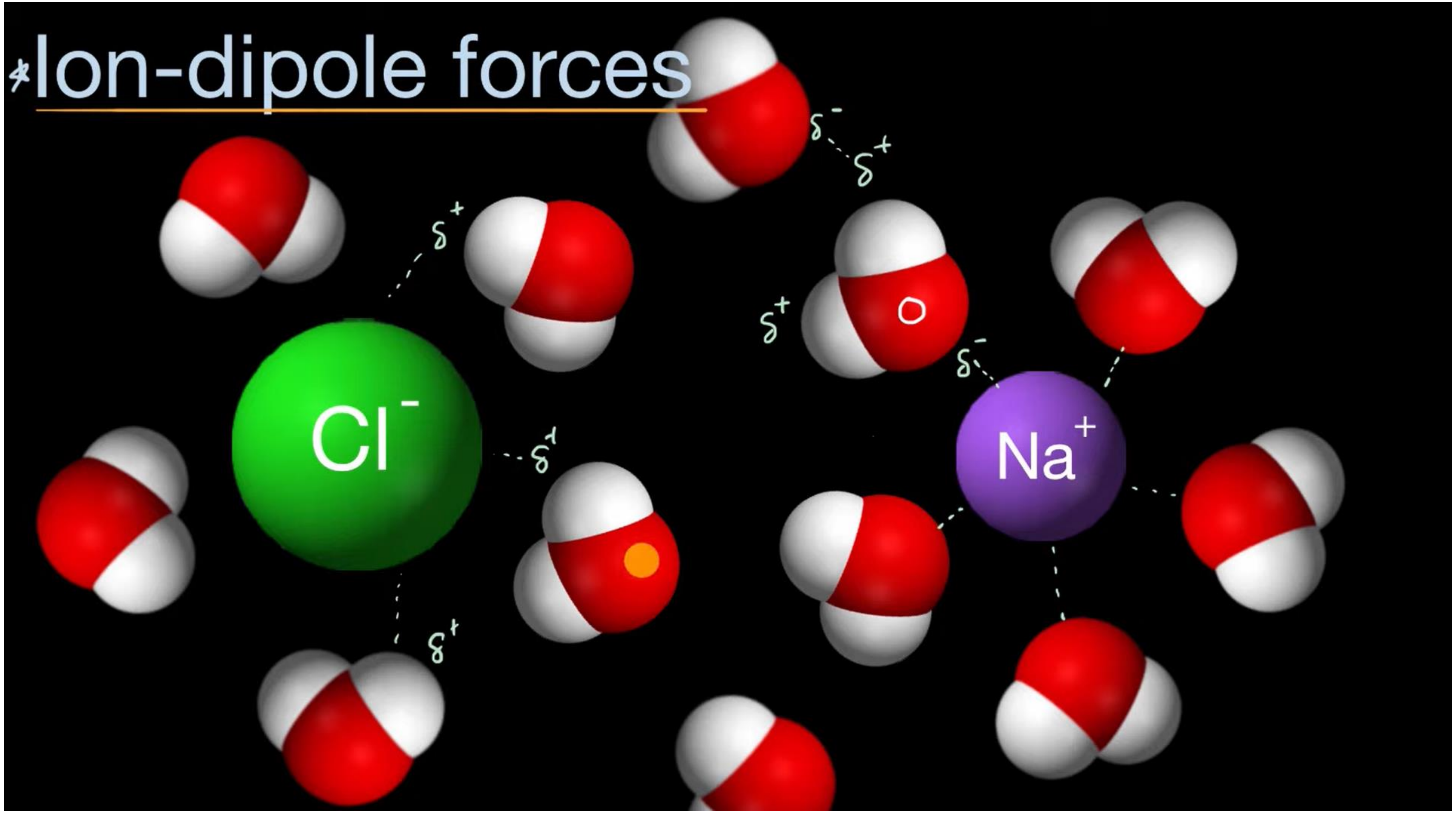


Hydrogen bonds

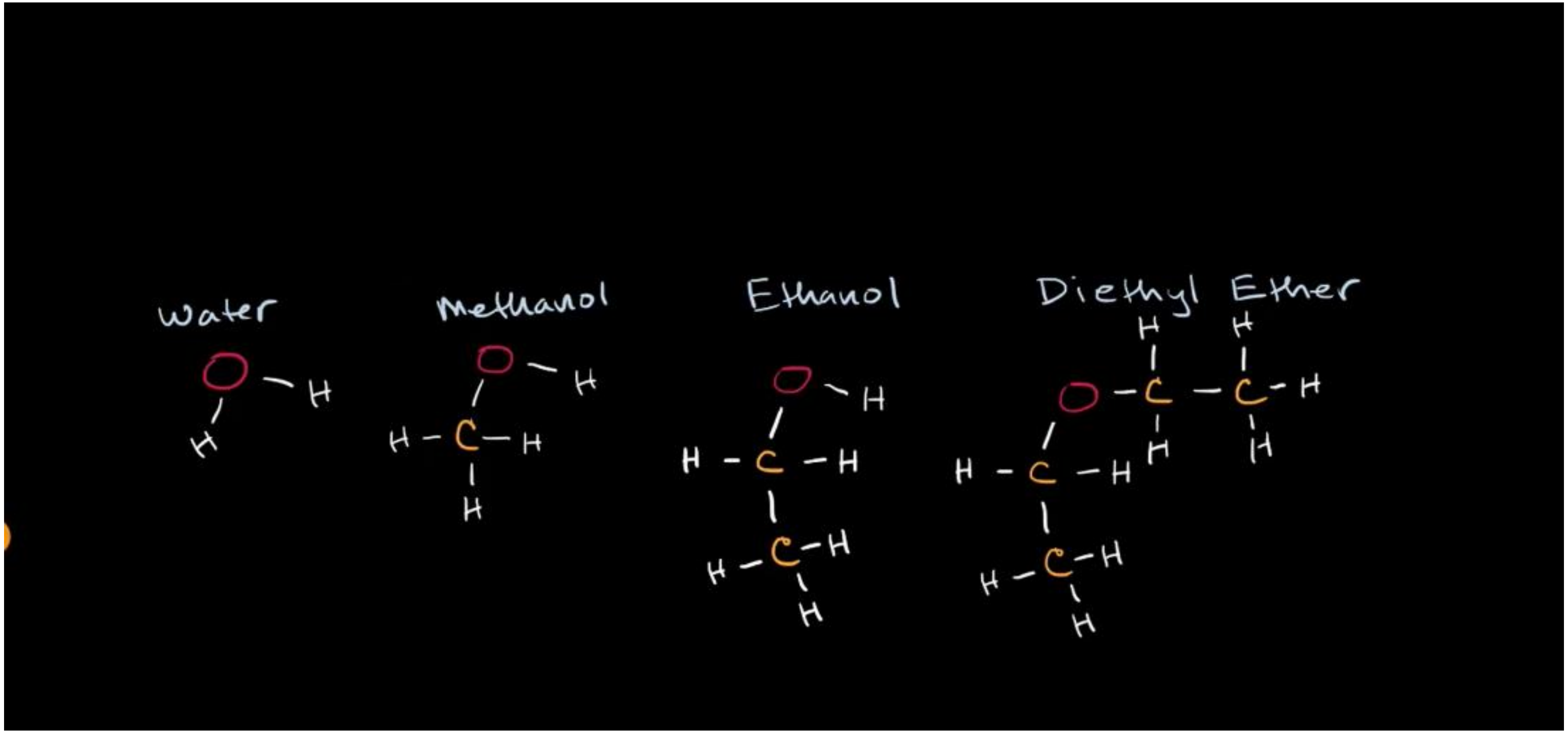



Hydrogen bonds


Thymine
Adenine
5′ end
3′ end
Phosphate-
deoxyribose
backbone
Guanine
Cytosine
3′ end
5′ end
image by Madeleine Price Ball

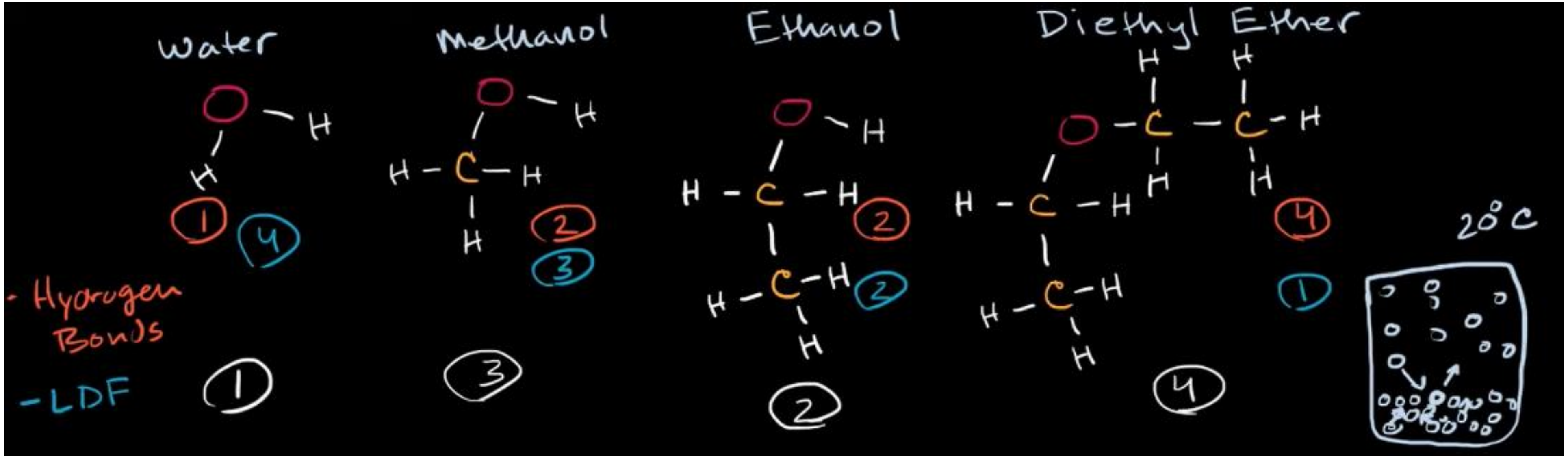

Ion-dipole forces
Cl⁻
Na⁺
δ⁻
δ⁺
o

Intermolecular Forces and Vapor Pressure



How would you order these molecules according to their BP?

Intermolecular Forces and Vapor Pressure



Name	Molecular formula	Molar mass (g/mol)	Boiling point (°C)	Vapor pressure at 20°C (mm Hg)
Water	H_2O	18.0	100.0	17.5
Methanol	CH_3OH	32.0	64.6	96.8
Ethanol	CH_3CH_2OH	46.1	78.3	43.5
Diethyl ether	$CH_3CH_2OCH_2CH_3$	74.1	34.5	439.5

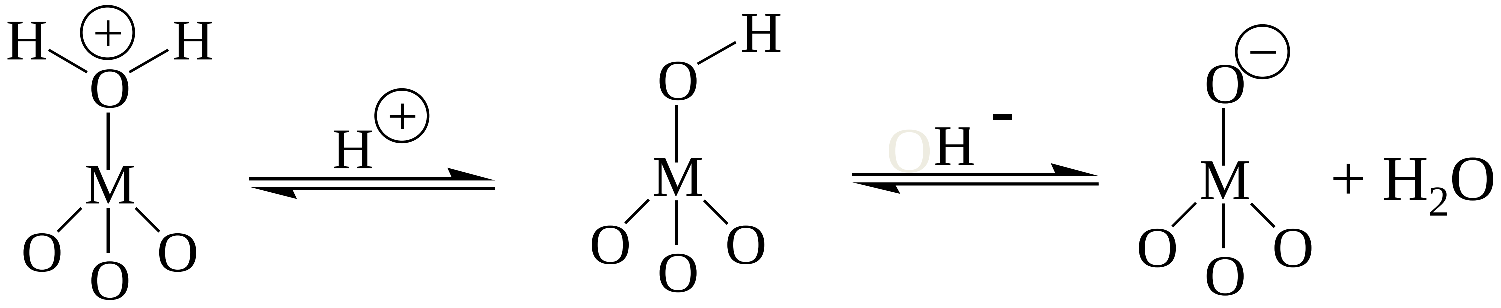
Electrostatic Forces &
The Electrical Double Layer

1) Sources of interfacial charge
2) Electrostatic theory: The electrical double layer
3) Electro-kinetic Phenomena
4) Electrostatic forces

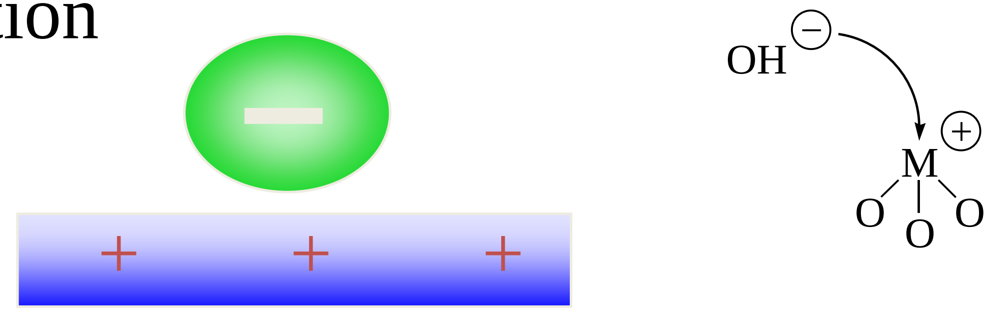
SOURCES OF INTERFACIAL CHARGE

- Immersion of some materials in an electrolyte solution. Two mechanisms can operate.

(1) Direct Ionization of surface groups.



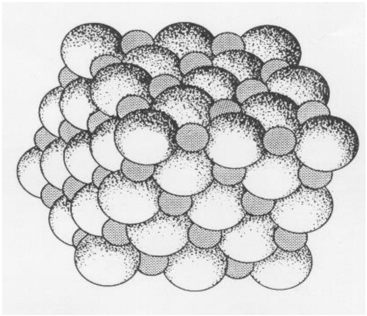
(2) Specific ion adsorption



SURFACE CHARGE GENERATION (cont.)

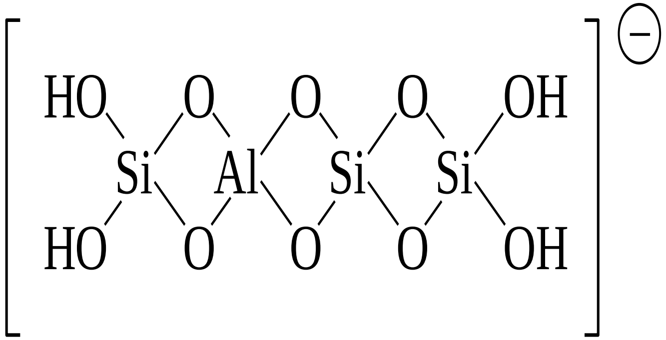
(3) Differential ion solubility

Some ionic crystals have a slight imbalance in number of lattice cations or anions on surface, *eg*. AgI, $BaSO_4$, CaF_2, NaCl, KCl

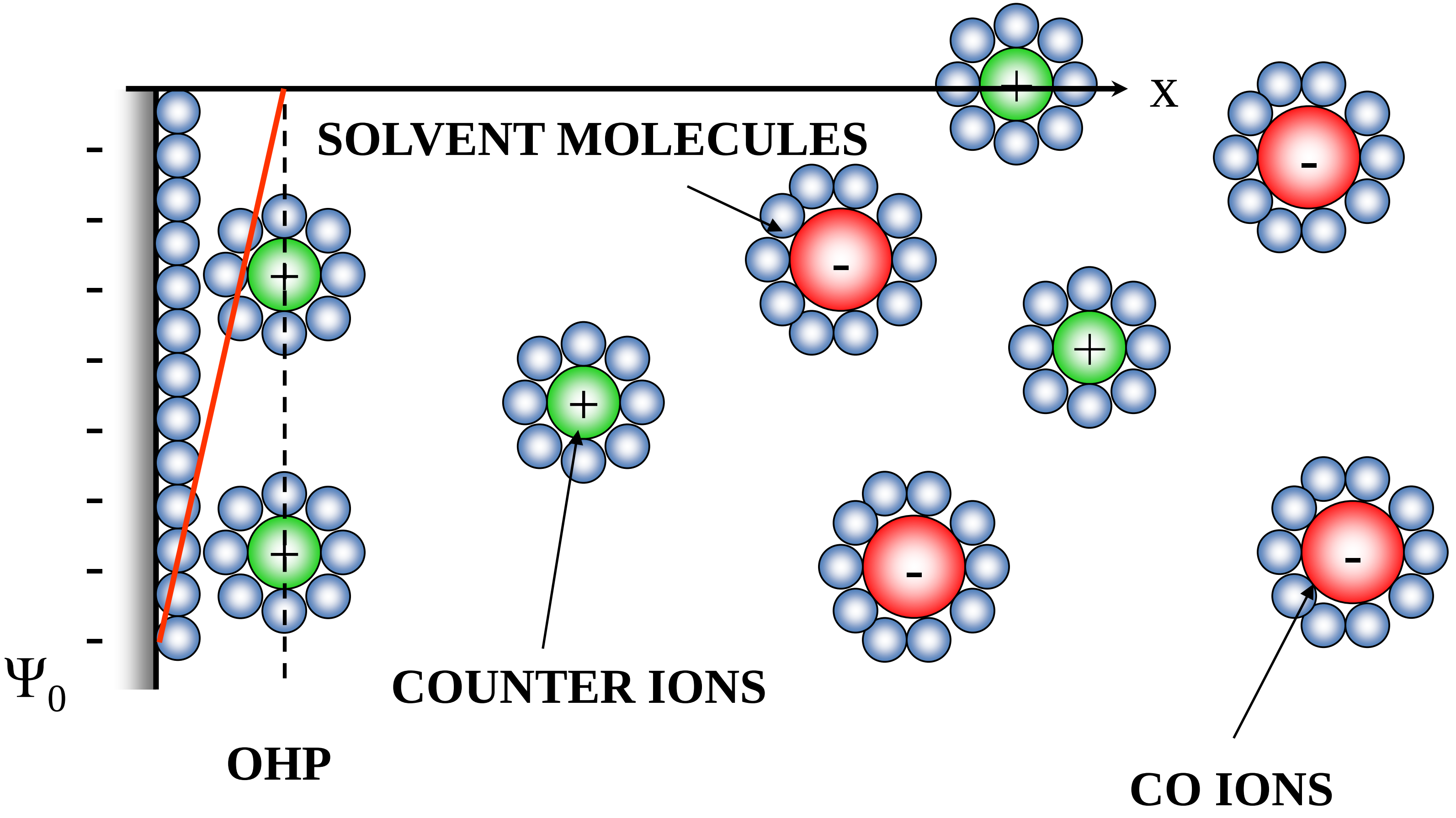


(4) Substitution of surface ions

eg. lattice substitution in kaolin



ELECTRICAL DOUBLE LAYERS



Helmholtz (100+ years ago) proposed that surface charge is balanced by a layer of oppositely charged ions

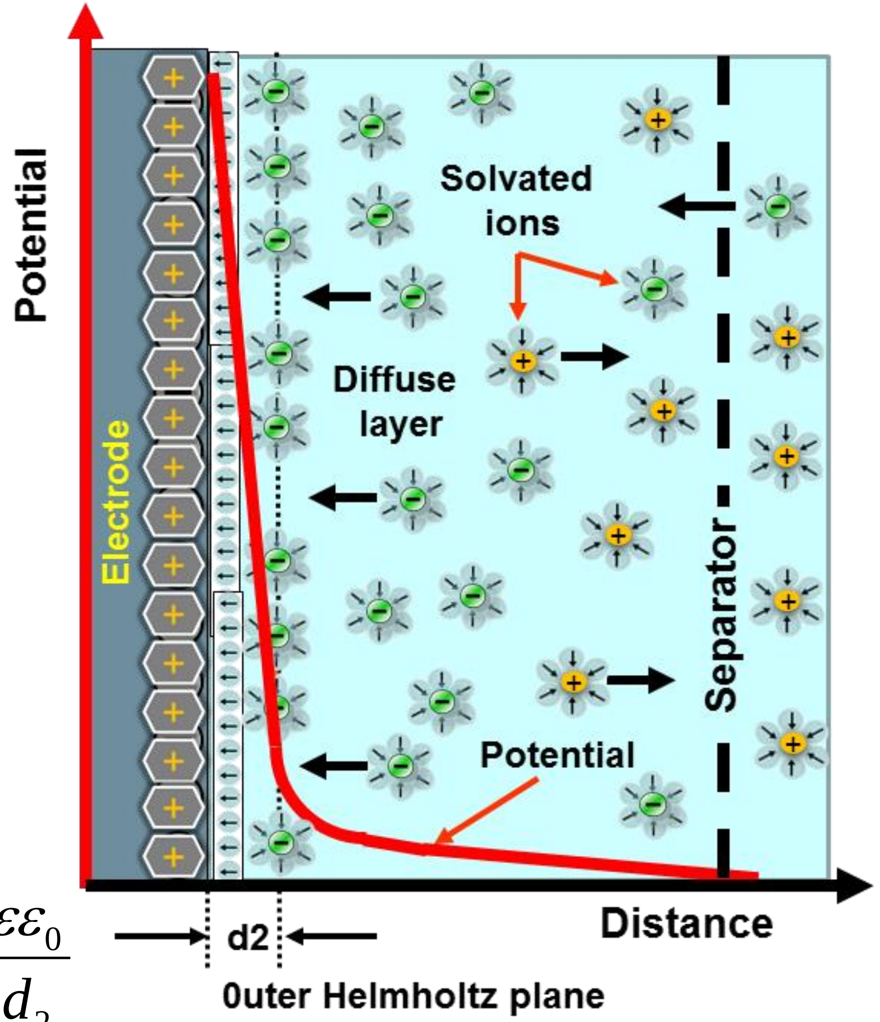
Helmholtz

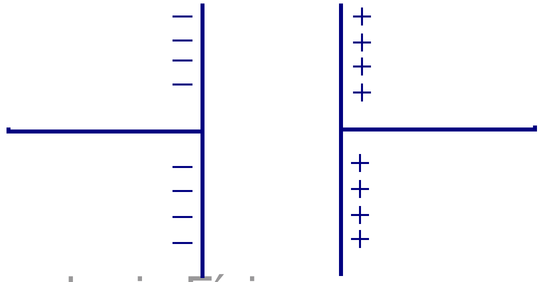
In 1853 Helmholtz showed that an electrical double layer (DL) is essentially a molecular dielectric and stores charge electrostatically.

This early model predicted a constant differential capacitance independent from the charge density depending on the dielectric constant of the electrolyte solvent and the thickness of the double-layer

The attracted ions are assumed to approach the electrode surface and form a layer balancing the electrode charge, the distance of approach is assumed to be limited to the radius of the ion and a single sphere of solvation round each ion. The overall result is two layers of charge (the double layer) and a potential drop which is confined to only this region (termed the outer Helmholtz Plane, OHP) in solution. The result is absolutely analogous to an electrical capacitor which has two plates of charge separated by some distance (d2)

Potential distribution in a Helmholtz double-layer





$$C_H = \frac{\partial \sigma}{\partial V} = \frac{\varepsilon \varepsilon_0}{d_2}$$

Gouy-Chapman Model (1910-1913)

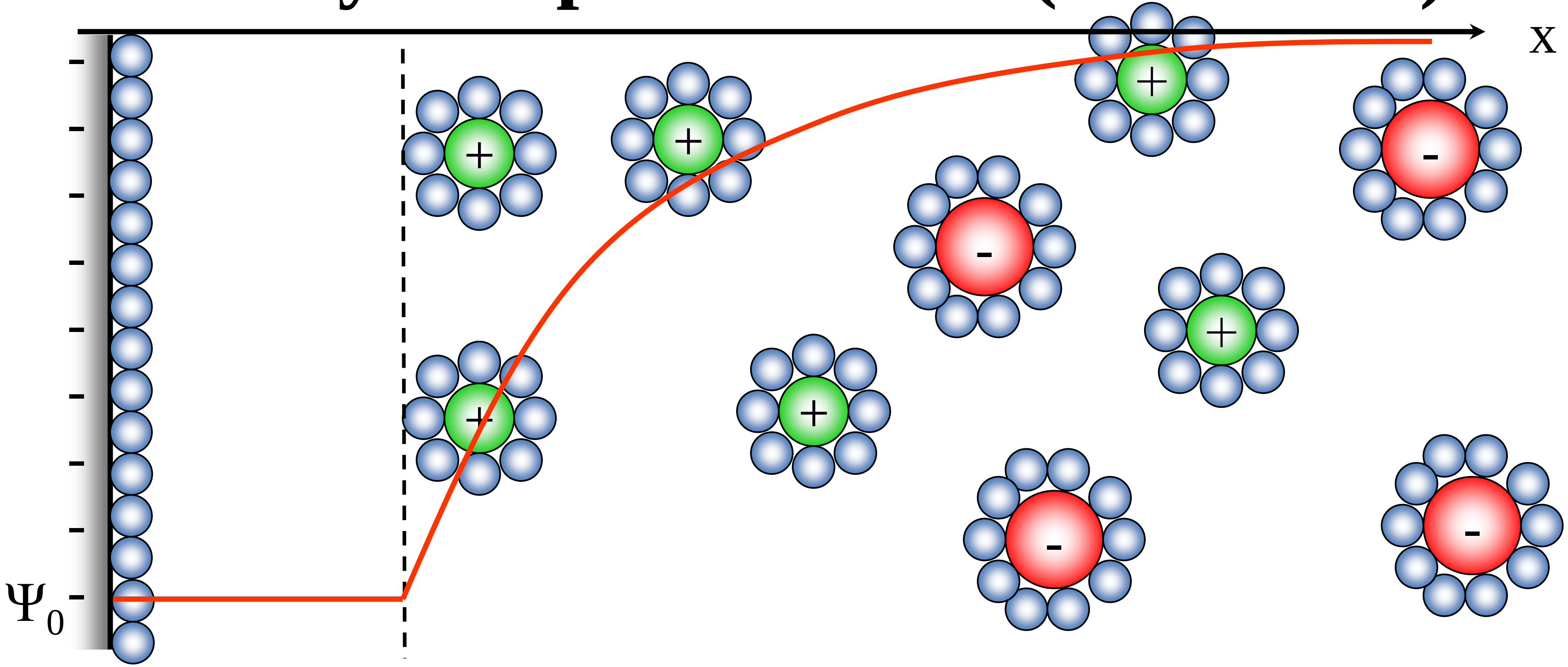
Gouy considered observations that capacitance was not a constant and that it depended on the applied potential and the ionic concentration. To account for this behaviour Gouy proposed that thermal motion kept the ions from accumulating on the surface of the electrode, instead forming a diffuse space charge

- **Observed that capacitance was not a constant and that it depended on the applied potential and the ionic concentration.**
- **The "Gouy-Chapman model" made significant improvements by introducing a diffuse model of the DL.**
- **In this model the charge distribution of ions as a function of distance from the metal surface allows Maxwell–Boltzmann statistics to be applied.**
- **Thus the electric potential decreases exponentially away from the surface of the fluid bulk**

$$C_G = \varepsilon\varepsilon_0 \kappa \cosh\frac{Z}{2}, \qquad \kappa = \sqrt{\frac{2ne^2Z^2}{\varepsilon\varepsilon_0 kT}}$$

Z is the valency, *n* is the number of ionic concentration, *T* is the absolute temperature, and *k* is the Boltzmann constant

Gouy-Chapman Model (1910-1913)



Diffusion plane

- Assumed Poisson-Boltzmann distribution of ions from surface
  - ions are point charges
  - ions do not interact with each other
- Assumed that diffuse layer begins at some distance from the surface

Stern (1924) / Grahame (1947) Model

Stern modified the Gouy-Chapman model by including a compact layer as well as Gouy's diffuse layer. The compact Stern layer consisted of a layer of specifically adsorbed ions. Grahame divided the Stern layer into two regions. He denoted the closest approach of the diffuse ions to the electrode surface as the outer Helmholtz plane (this is sometimes referred to as the Gouy plane). The layer of adsorbed ions at the electrode surface was designated as being the inner Helmholtz plane:

$$\frac{1}{C} = \frac{1}{C_H} + \frac{1}{C_G}$$

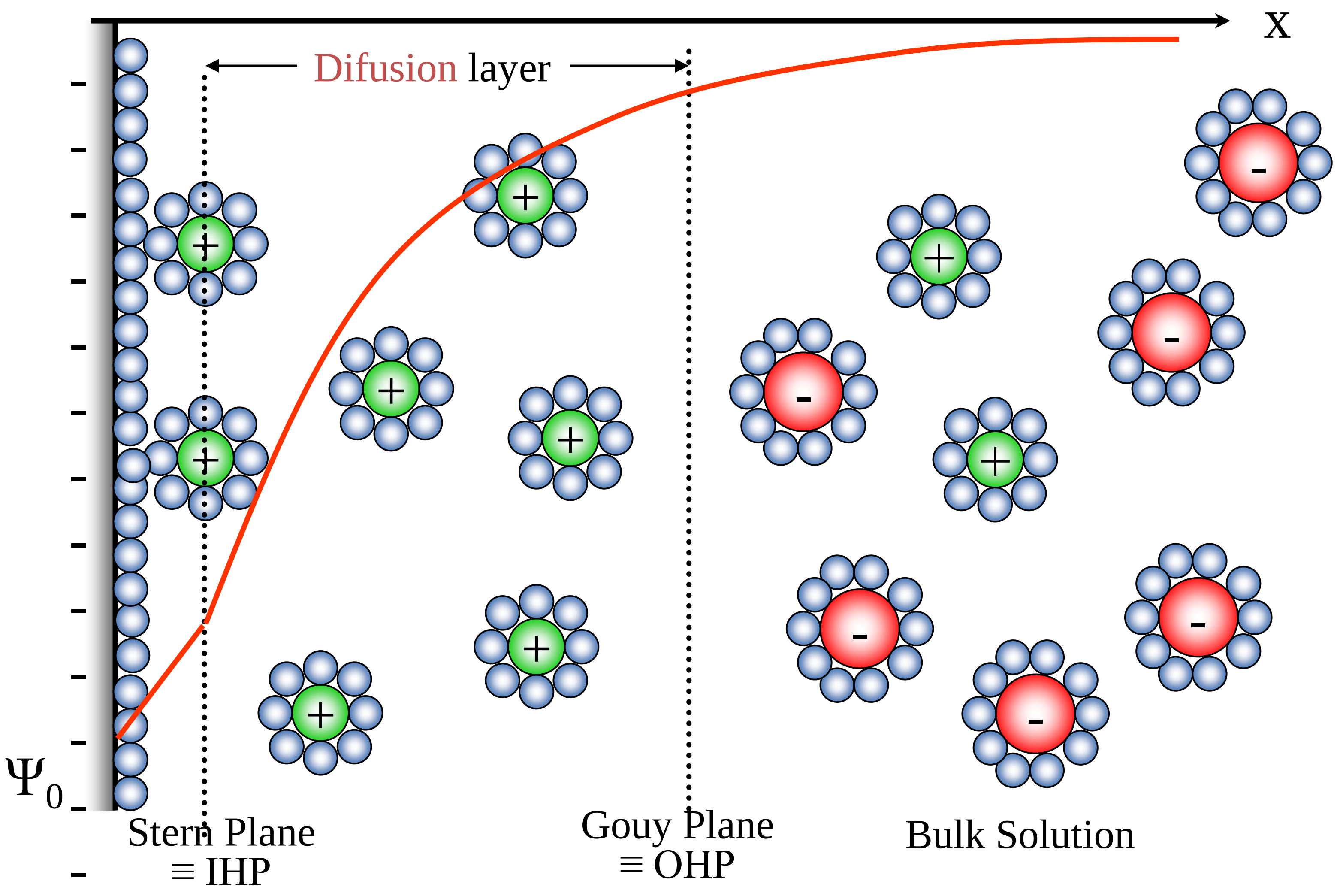
Which becomes invalid if specific adsorption of ions occurs. In that case:

$$\frac{1}{C} = \frac{1}{C_H} + \frac{1}{C_G}\left(1 + \frac{\partial \sigma_A}{\partial \sigma}\right)$$

σ is charge density on the electrode, and σ_A is the surface charge of the adsorbed ions

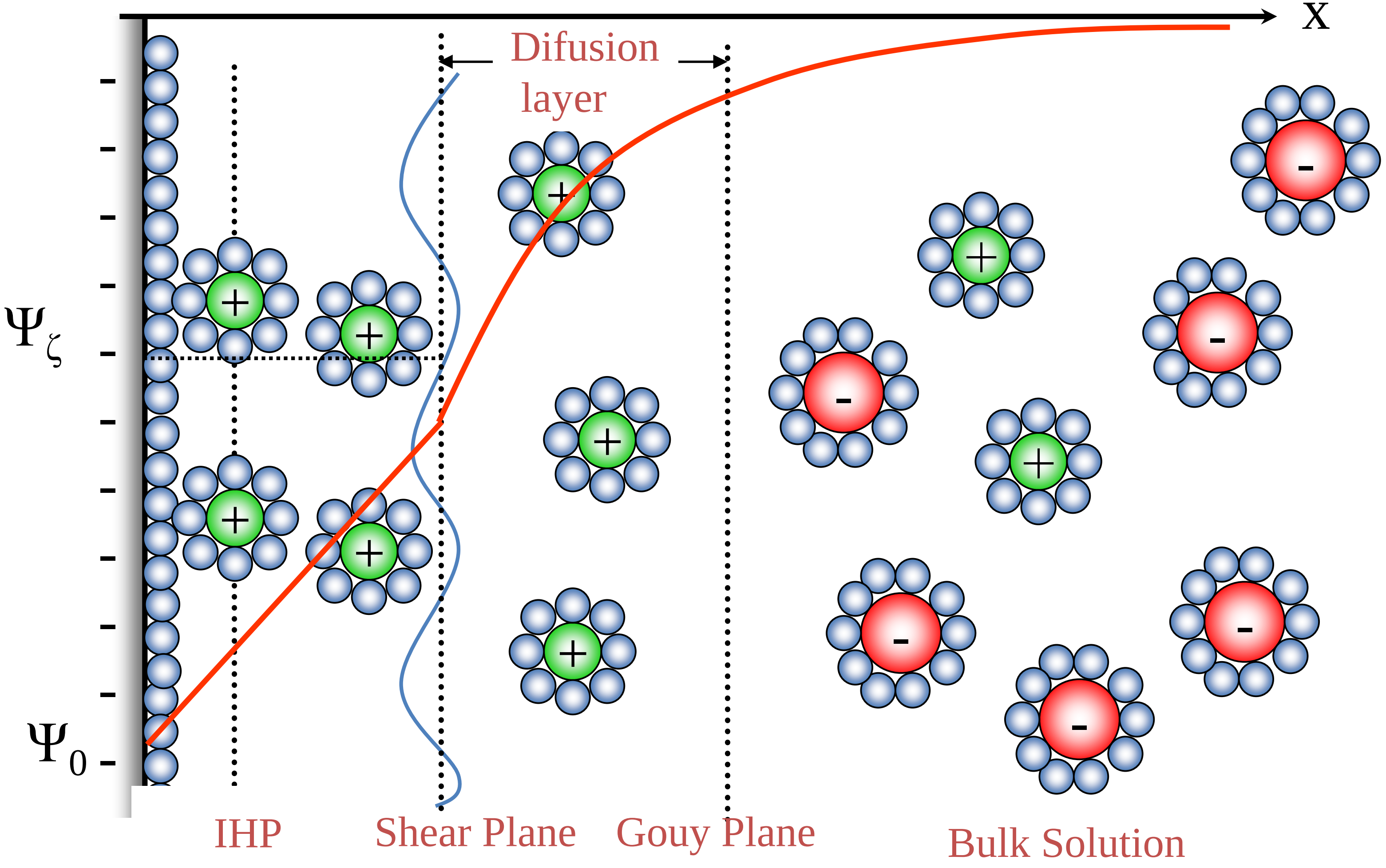
Stern (1924) / Grahame (1947) Model

Gouy/Chapman diffuse double layer and layer of adsorbed charge.
Linear decay until the Stern plane.



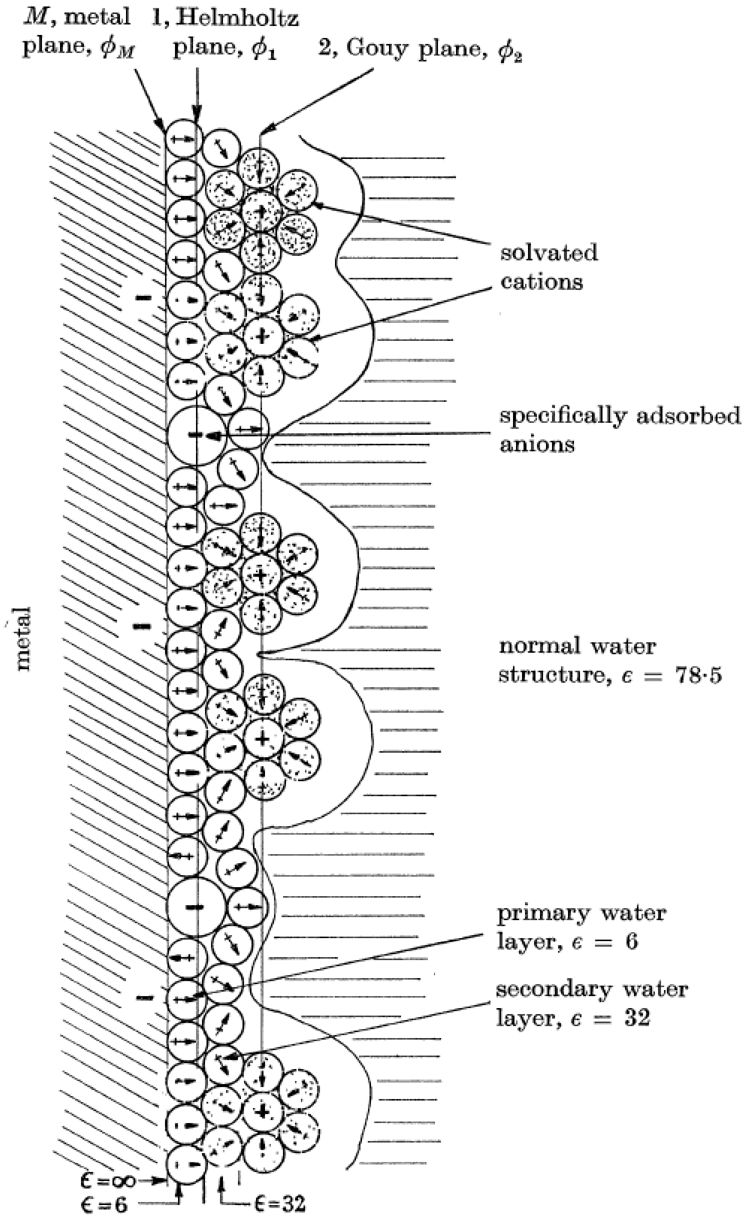
Stern (1924) / Grahame (1947) Model

In different approaches the linear decay is assumed to be until the shear plane, since there is the barrier where the charges are considered static.



Bockris, Devanathan and Müller Model (1963)

Bockris, Devanathan and Muller proposed a model that included the action of the solvent. They suggested that a layer of water was present within the inner Helmholtz plane at the surface of the electrode. The dipoles of these molecules would have a fixed alignment because of the charge in the electrode. Some of the water molecules would be displaced by specifically adsorbed ions. Other layers of water would follow the first, but the dipoles in these layers would not be as fixed as those in the first layer.



POISSON-BOLTZMANN DISTRIBUTION

1st Maxwell law (Gauss law): "*The total of the electric flux out of a closed surface is equal to the charge enclosed divided by the permittivity*"

$$\nabla \cdot E = \frac{\rho(r)}{\varepsilon_0 \varepsilon_r}$$

Electric field is the differential of the electric potential

$$E(r) = -\nabla \cdot \Psi(r)$$

Combining the two equations we get:

$$\nabla^2 \cdot \Psi(r) = -\frac{\rho(r)}{\varepsilon_0 \varepsilon_r}$$

Which for one dimension becomes:

$$\frac{d^2\Psi(x)}{dx^2} = -\frac{\rho(x)}{\varepsilon\varepsilon_0}$$

Definitions

E: Electric filed

Ψ: Electric potential

ρ: Charge density

E_Q: Energy of the ion

x, *r* : Distance

Boltzmann ion distribution

$$\rho(x) = \rho_0 e^{-\frac{E_Q}{kT}} = n_0 Zq e^{-\frac{qZ\Psi}{kT}}$$

Assuming Boltzmann ion distribution:

$$\frac{d^2\Psi(x)}{dx^2} = -\frac{\rho(x)}{\varepsilon\varepsilon_0} = -\frac{1}{\varepsilon\varepsilon_0}\sum_i^n Z_i n_i q e^{-Z_i q\Psi/kT} = -\frac{Zqn}{\varepsilon\varepsilon_0}\left(e^{-Zq\Psi/kT} - e^{Zq\Psi/kT}\right)$$

For a symmetric electrolyte

POISSON-BOLTZMANN DISTRIBUTION

$$\frac{d^2\psi}{dx^2} = \frac{2Zqn}{\varepsilon_r \varepsilon_0} \sinh\left(\frac{Zq\psi}{kT}\right)$$

Z = electrolyte valence,
q = elementary charge (C)
n = electrolyte concentration(#/m^3)
ε_r = dielectric constant of medium
ε_0 = permittivity of a vacuum (F/m)
k = Boltzmann constant (J/K)
T = temperature (K)

- Poisson-Boltzmann distribution describes the EDL
- Defines potential as a function of distance from a surface
    - ions are point charges
    - ions do not interact with each other

POISSON-BOLTZMANN DISTRIBUTION
Debye-Hückel approximation

For $\dfrac{Zq\Psi}{kT} << 1$ then:

$$\frac{d^2\psi}{dx^2} = \frac{2Zqn}{\varepsilon_r \varepsilon_0} \sinh\left(\frac{Zq\psi}{kT}\right) \approx \frac{2Zqn}{\varepsilon_r \varepsilon_0} \frac{Zq\psi}{kT} = \frac{2n(Zq)^2}{\varepsilon_r \varepsilon_0 kT} \Psi(x) = \kappa^2 \Psi(x)$$

The solution is a simple exponential decay (assuming $\Psi(0)=\Psi_0$ and $\Psi(\infty)=0$):

$$\Psi(x) = \Psi_0 e^{-\kappa x}$$

Debye-Hückel parameter (κ) describes the decay length

$$\boxed{\kappa = \sqrt{\frac{2n(Zq)^2}{\varepsilon_r \varepsilon_0 kT}}}$$

DOUBLE LAYER FOR MULTIVALENT ELECTROLYTE: DEBYE LENGTH

Debye-Hückel parameter (κ) describes the decay length

$$\kappa = \left(\frac{q^2}{\varepsilon_r \varepsilon_0 kT} \sum_{i=1}^{n} C_i Z_i^2 \right)^{1/2}$$

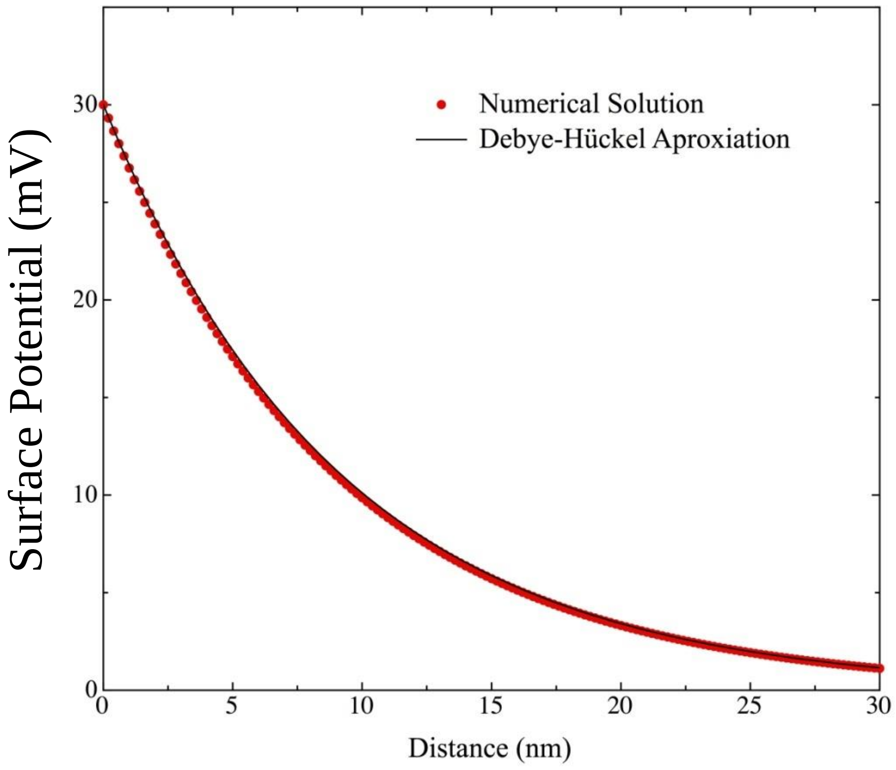
Z_i	= electrolyte valence
q	= elementary charge (C)
C_i	= ion concentration ($\#/m^3$)
n	= number of ions
ε_r	= dielectric constant of medium
ε_0	= permittivity of a vacuum (F/m)
k	= Boltzmann constant (J/K
T	= temperature (K)

κ^{-1} (Debye length) has units of length

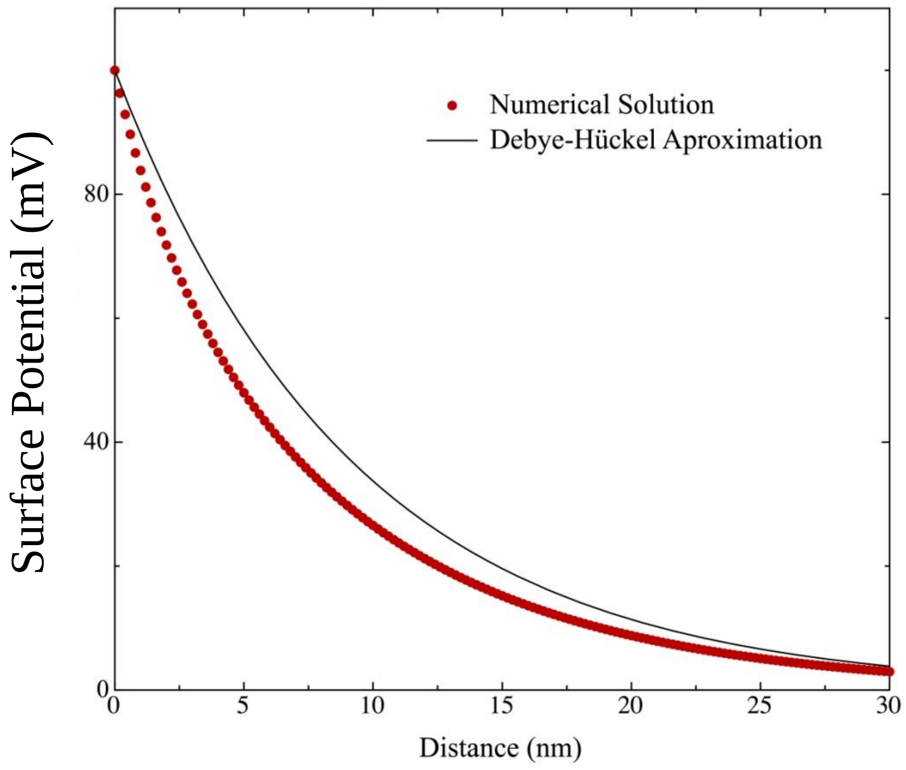
POISSON-BOLTZMANN DISTRIBUTION

Exact Solution

For 0.001 M 1-1 electrolyte

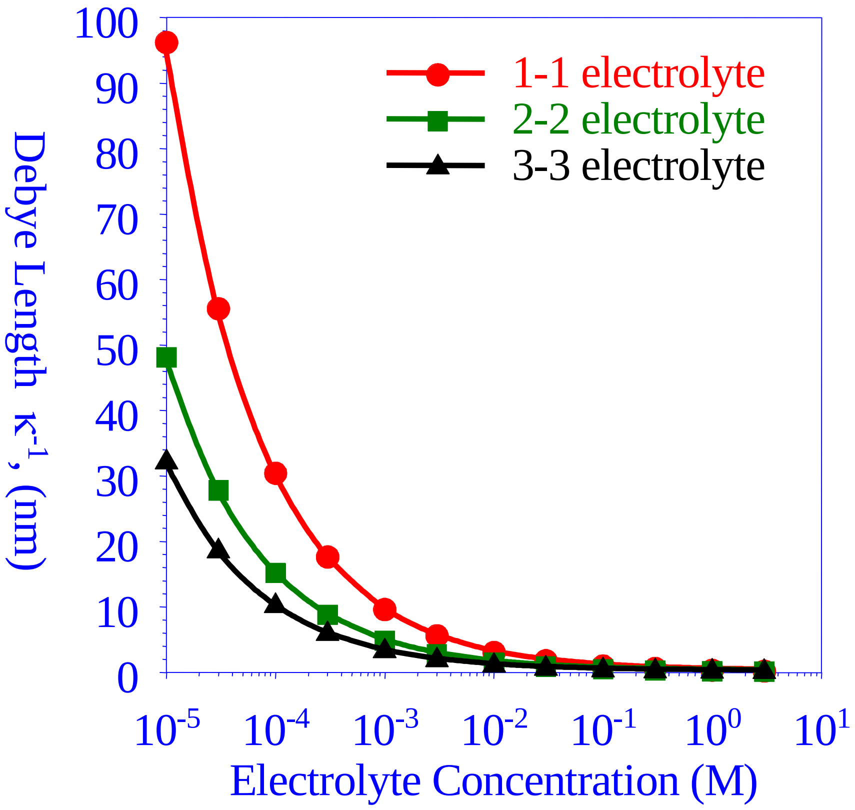


$$\frac{Ze\Psi}{kT} < 1$$



$$\frac{Ze\Psi}{kT} > 1$$

DEBYE LENGTH AND VALENCY



Ions of higher valence are more effective in screening surface charge.

Outline

- What is a biosensor?
- Biosensors with electronic transduction
  - Field-effect transistors
  - Electrochemical sensors
- Solid-liquid interfaces: the electrical double layer
- Techniques to study the solid-liquid interface:
  - The quartz crystal microbalance
  - The Langmuir isotherm
  - Electrokinetic phenomena and the Zeta potential measurement
  - DLVO theory
- 2D materials and interfaces
- Biosensor development and fabrication

Liquid-Solid Interface

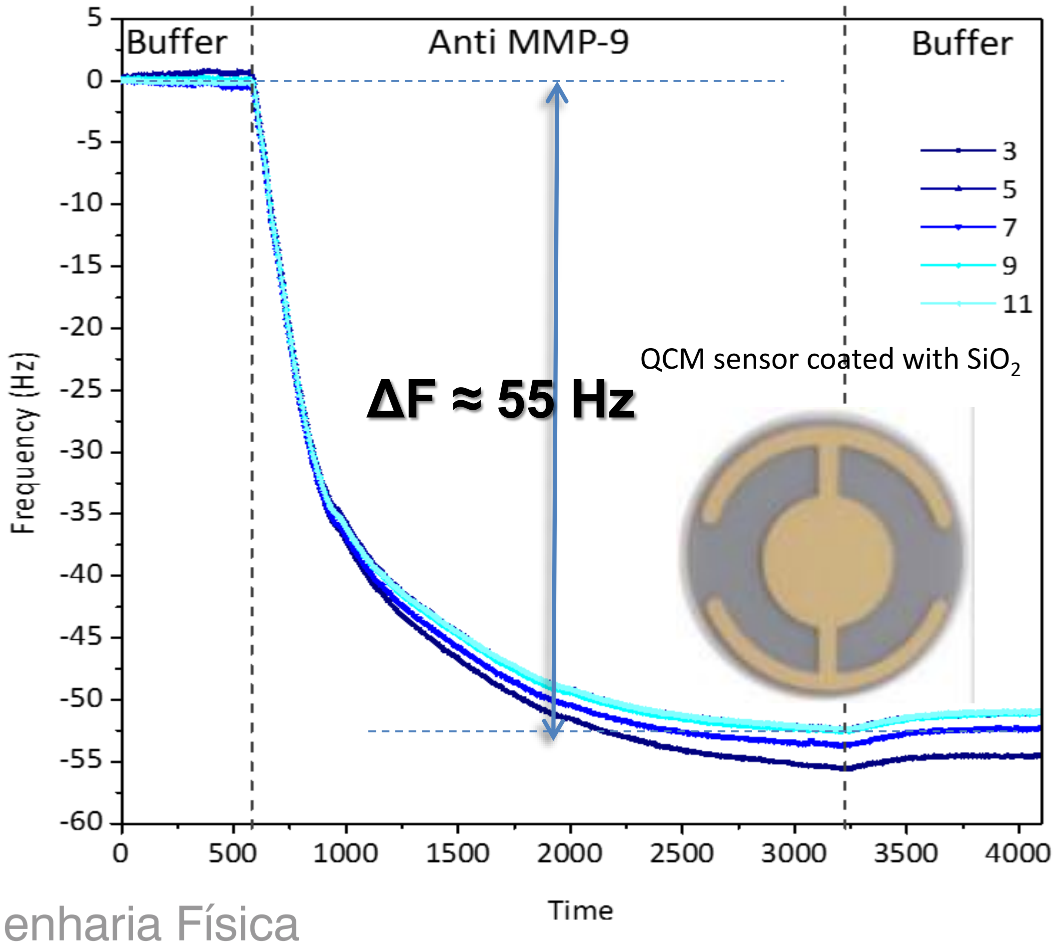
- Separation techniques such as :
- column chromatography, HPLC, Paper Chromatography, TLC
- They are examples of the adsorption of solutes at the liquid-solid interface.
- Liquid-solid interfaces can be found everywhere and in any form and size, from electrode surfaces to ship hulls.

Liquid-Solid Interface; Colloids

- Not so obvious examples for a solid-liquid interface is a colloid particle.
- Colloids are widely spread in daily use:
- Paint
- Blood
- Air pollution

Liquid-Solid Interface

- A very sensitive method to measure the amount of adsorbed material is by using a QCM (quartz crystal microbalance)



QCM

fundamental frequency ≈ 4.95 MHz

6 odd overtones (n = 3, 5, 7, 9, 11, 13)

- baseline in the blank 10 mM PB
- Antibody (Anti MMP-9) was immobilized by recirculating 1 ml of 20 µg/ml solution in the same buffer through the QCM flow cell

Quartz crystal microbalance

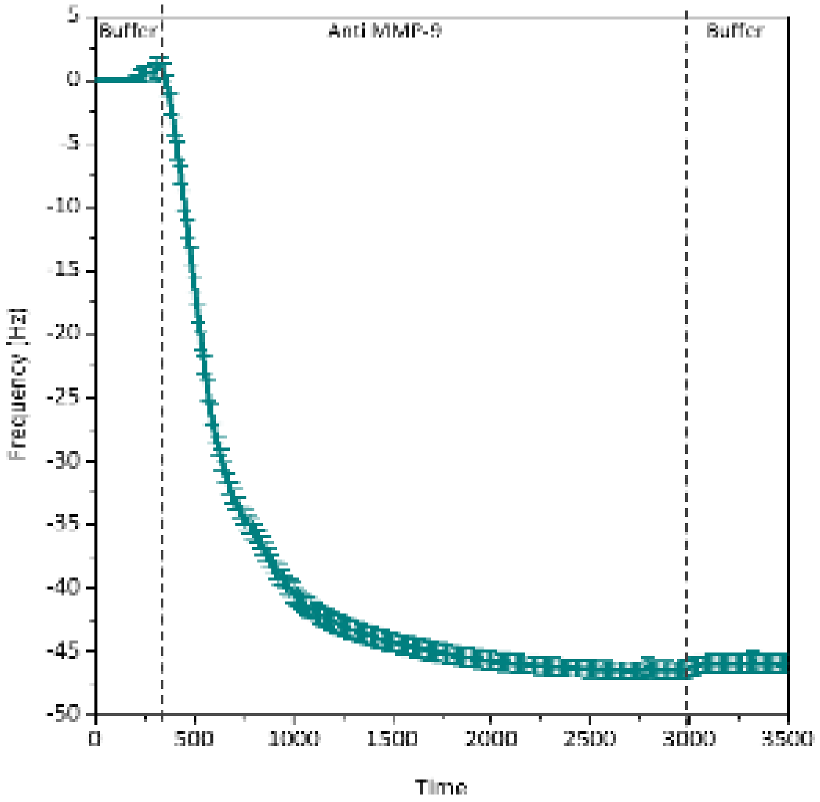
Surface coverage estimation based on QCM signal:

Sauerbrey equation: $\Delta m = -C_{QCM} \times \Delta f$

(with correction for molecular hydration)

C_{QCM} = 17.7 ng·cm^{-2} ·Hz^{-1}

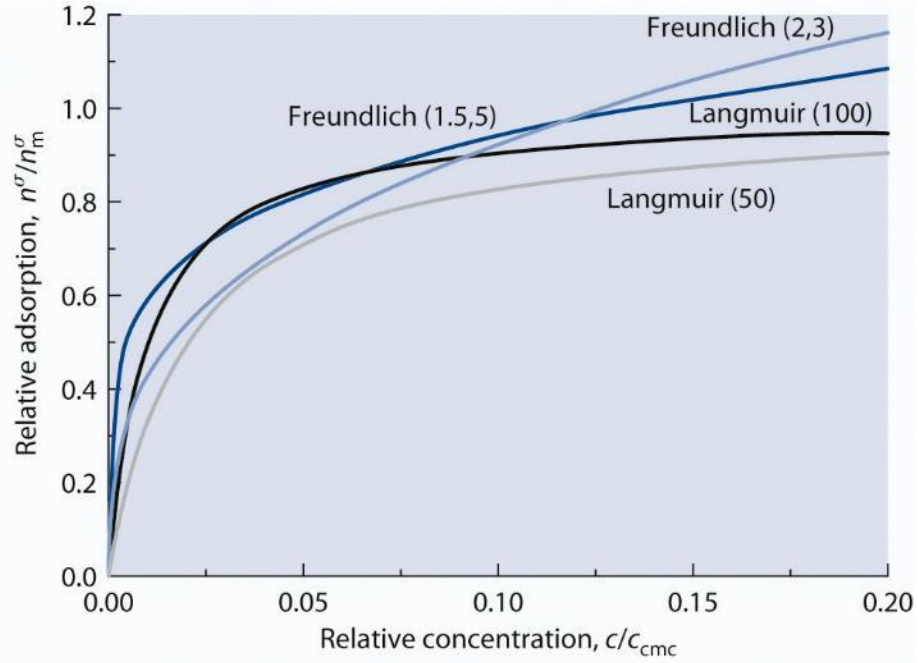
S. Ray et al., Langmuir. 31 (2015) 1921-1930



N ≈ 1.74 x 10^{12} Antibodies.cm^{-2}

Liquid-Solid Interface

- Adsorption at low solute concentration:
- These isotherms can be either fitted by the langmuir isotherm
- Or the freundlich isotherm



The Langmuir Isotherm

$$A_g + S \rightleftharpoons A_{ad} \quad , \quad A_g - \text{adsorbate molecule} \; ; \; S - \text{empty site} \; ; \; A_{ad} - \text{adsorbed complex}$$

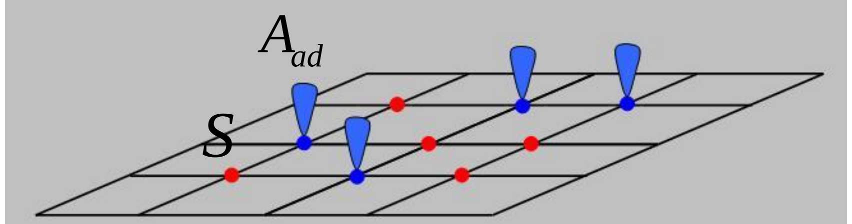
$$\theta_A = \frac{V}{V_m} = \frac{K_{eq}^A}{1 + K_{eq}^A n_A}$$

θ_A – fraccional occupancy of adsorption sites

V_m – volume of the monolayer

n_A – concentration of the adsorbate in solution

K_{eq} – reaction equilibrium constant



The adsorption sites (heavy dots) are equivalent and can have unit occupancy. Also, the adsorbates are immobile on the surface.

The Langmuir Isotherm

$$\boxed{A + S \rightleftharpoons A_{ad}}$$

r_{ad} – rate of adsorption $\qquad r_{ad} = \kappa_{ad}[A][S]$

r_d – rate of desorption $\qquad r_d = \kappa_d[A_{ad}]$

$[A]$ – concentration of A above the surface

$[S]$ – concentration of free sites / m^2

$[A_{ad}]$ – concentration of A adsorbed / m^2

κ_{ad} , κ_d - constants of forward adsorption and backward desorption reaction

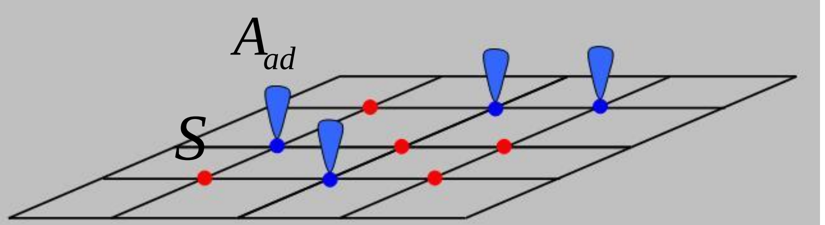
At equilibirum $r_d = r_{ad}$, therefore:

$$\frac{[A_{ad}]}{[A][S]} = \frac{\kappa_{ad}}{\kappa_d} = K_{eq}^A$$

$$[S_0] = S_0/a \quad \text{and} \quad [S_0] = [S] + [A_{ad}]$$

$$[S_0] = \frac{[A_{ad}]}{[A]K_{eq}^A} + [A_{ad}] = \frac{1+[A]K_{eq}^A}{[A]K_{eq}^A}[A_{ad}]$$

$$\theta_A = \frac{[A_{ad}]}{[S_0]} \Rightarrow \boxed{\theta_A = \frac{[A]K_{eq}^A}{1+[A]K_{eq}^A}} \qquad c.q.d.$$



The adsorption sites (heavy dots) are equivalent and can have unit occupancy. Also, the adsorbates are immobile on the surface.

The Freundlich isotherm

$$\boxed{\frac{x}{m} = Kc^{1/n} \Leftrightarrow \log\frac{x}{m} = \log K + \frac{1}{n}\log c}$$

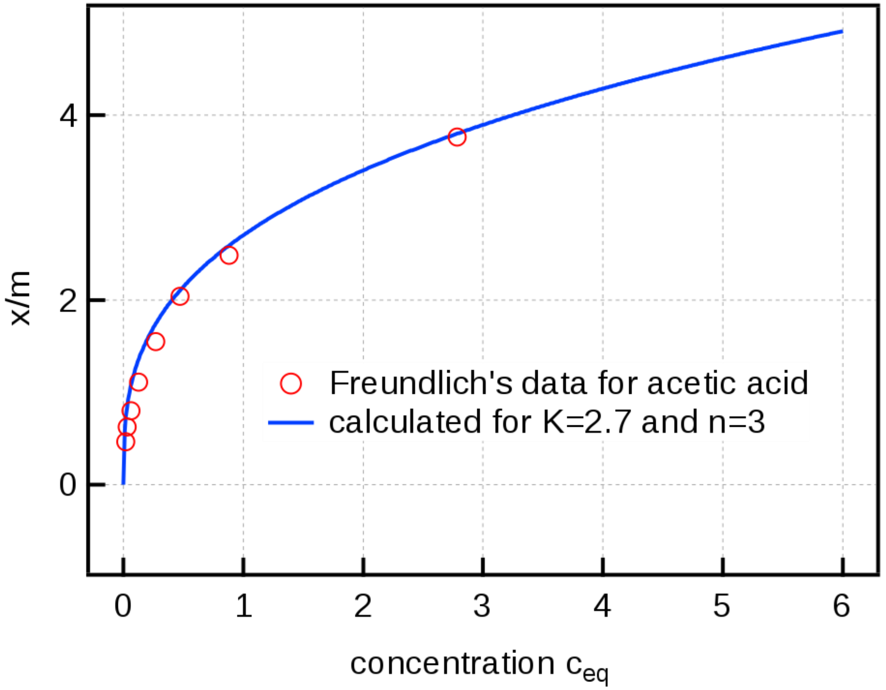
x – masse of adsorbate

m – mass of adsorbent

c = equilibrium concentration of adsorbate insolution

K and n are constants for a given adsorbate and adsorbent at a particular temperature

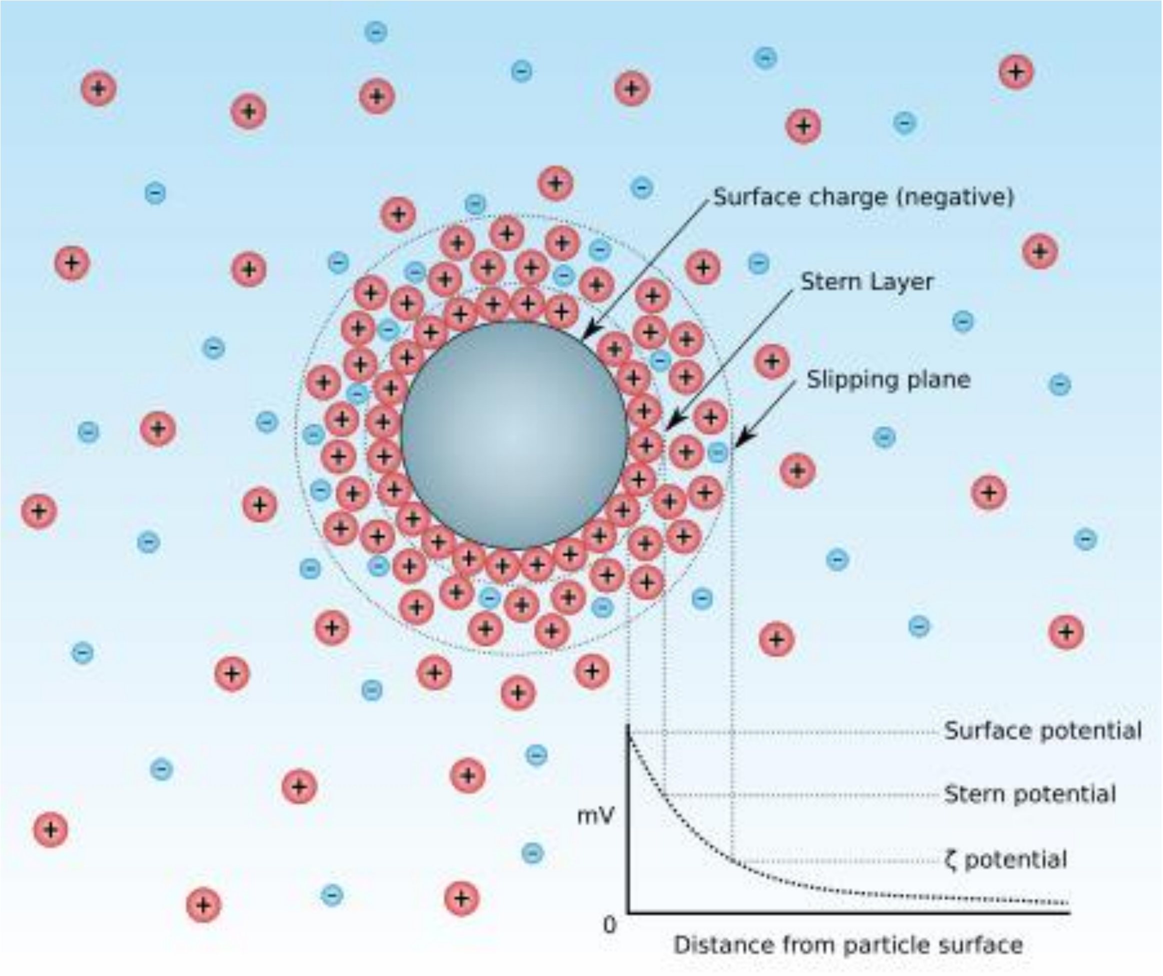
At high pressure 1/n = 0, hence extent of adsorption becomes independent of concentration



Liquid-Solid Interface; Colloids

- Not so obvious examples for a solid-liquid interface is a colloid particle.
- Colloids are widely spread in daily use:
- Paint
- Blood
- Air pollution

The Zeta potential

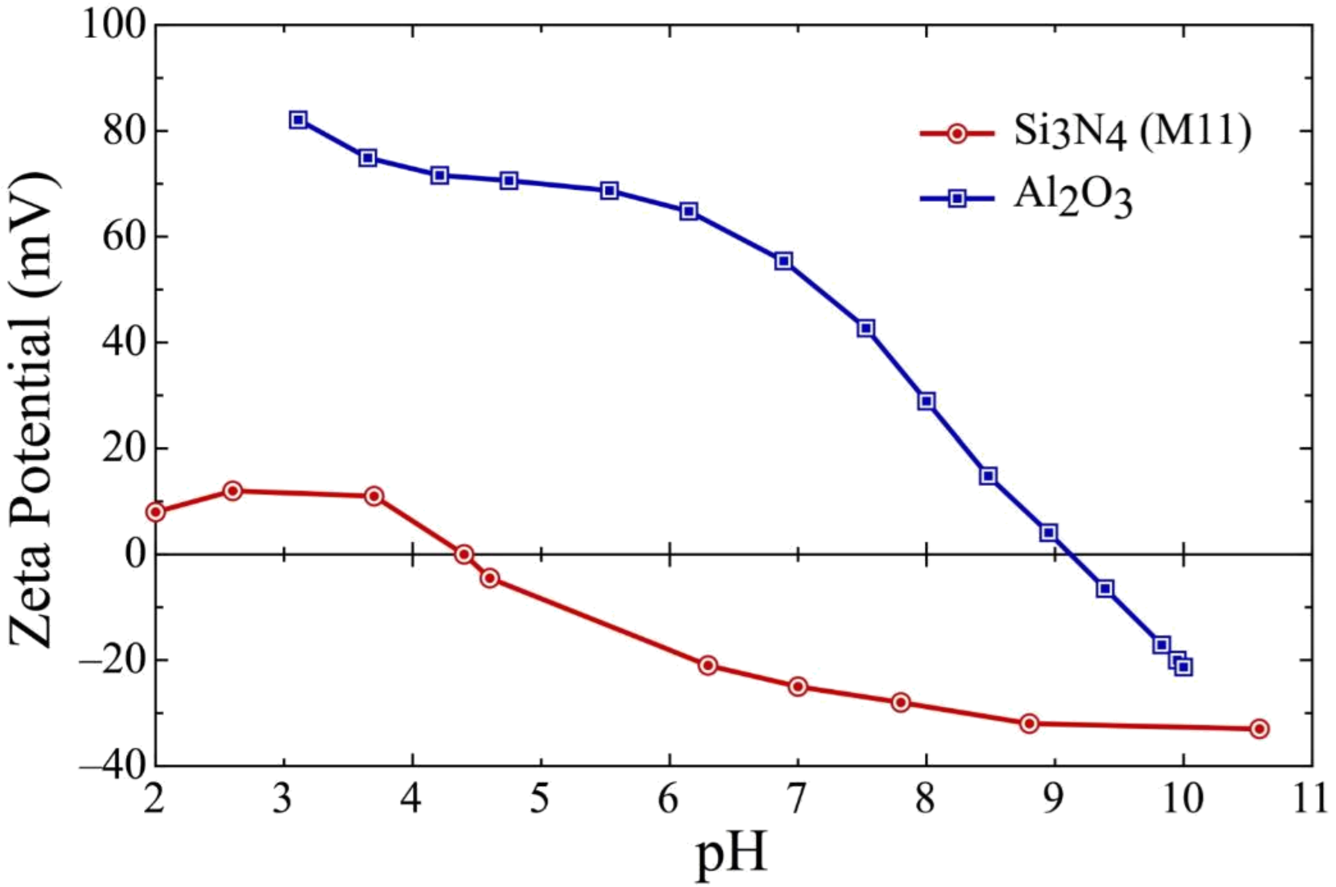


The Zeta potential

Zeta potential (mV)	Stability behavior
0 to ±5	Rapid coagulation or flocculation
±10 to ±30	Incipient instability
±30 to ±40	Moderate stability
±40 to ±60	Good stability
>61	Excellent stability

doi:10.1016/B978-0-08-100557-6.00003-1

Zeta potential

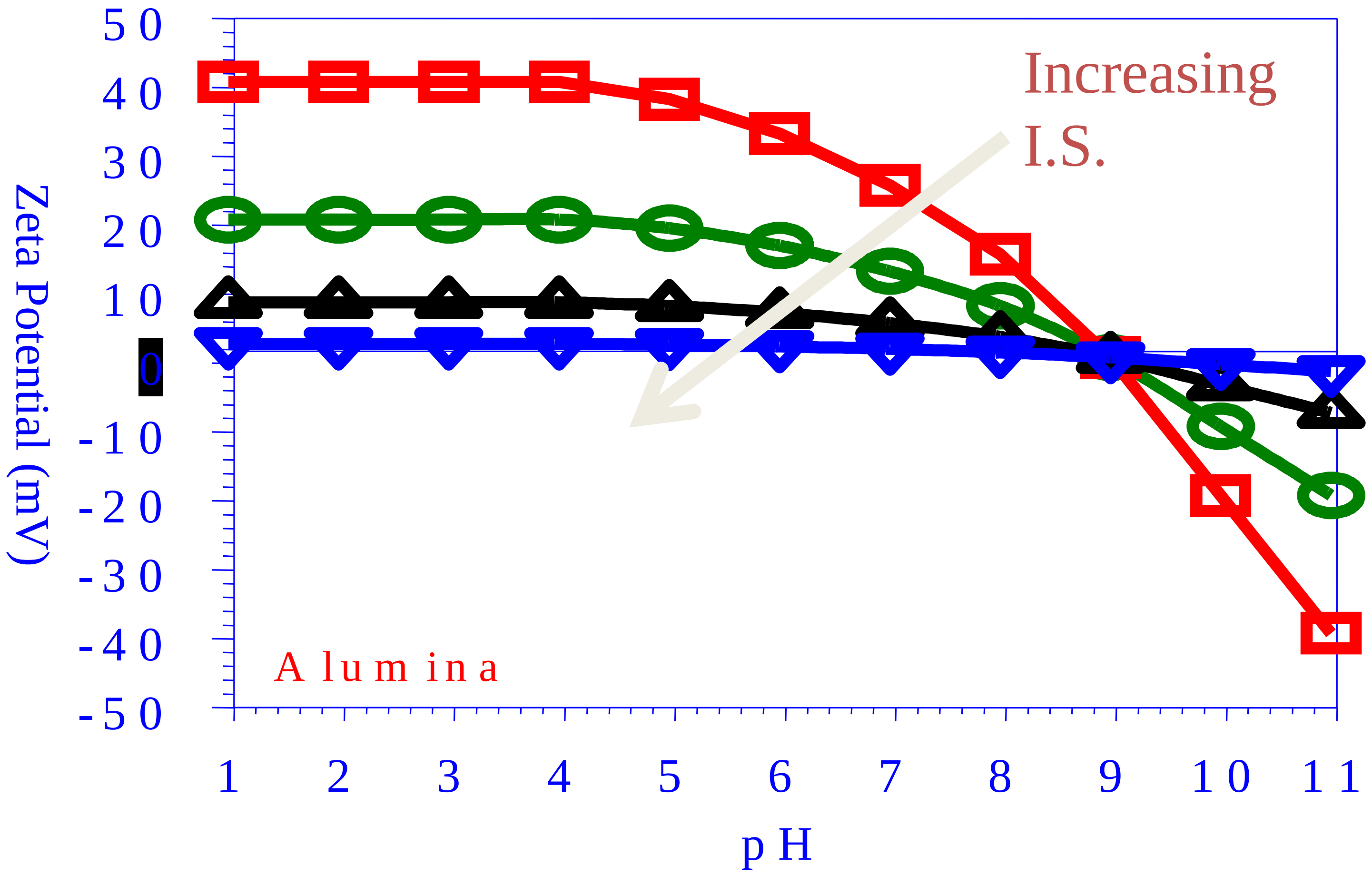


Point of Zero Charge (PZC) - pH at which surface potential = 0
Isoelectric Point (IEP) - pH at which zeta potential = 0

Question: What will happen to a mixed suspension of Alumina and Si_3N_4 particles in water at pH 4, 7 and 9?

Zeta potential

Effect of Ionic Strength



VISCOSITY

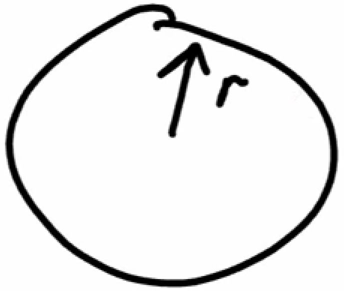
Viscosity, η

@ 25°C, 1 bar

	material	η (mPa·s)
Water	H_2O	0.89
Ethylene-glicol		16
Glicerol		950

Stokes Equation

$$\vec{F}_D = -6\pi\eta r\vec{v}$$



laminar

ZETA POTENTIAL MEASUREMENT

- **Electrophoresis** - ζ determined by the rate of diffusion (electrophoretic mobility) of a charged particle in an applied DC electric field.
- **PCS** - ζ determined by diffusion of particles as measured by photon correlation spectroscopy (PCS) in applied field
- **Acoustophoresis** - ζ determined by the potential created by a particle vibrating in its double layer due to an acoustic wave
- **Streaming Potential** - ζ determined by measuring the potential created as a fluid moves past macroscopic surfaces or a porous plug

ZETA POTENTIAL MEASUREMENT

Electrophoresis

Smoluchowski Formula (1921)

assumed $\kappa a >> 1$ κ = Debye parameter

a = particle radius

- electrical double layer thickness much smaller than particle

$$v = \frac{\varepsilon_r \varepsilon_0 \zeta}{\eta} E$$

$$\mu_E = \frac{\varepsilon_r \varepsilon_0 \zeta}{\eta}$$

v = velocity,

ε_r = media dielectric constant

ε_0 = permittivity of free space

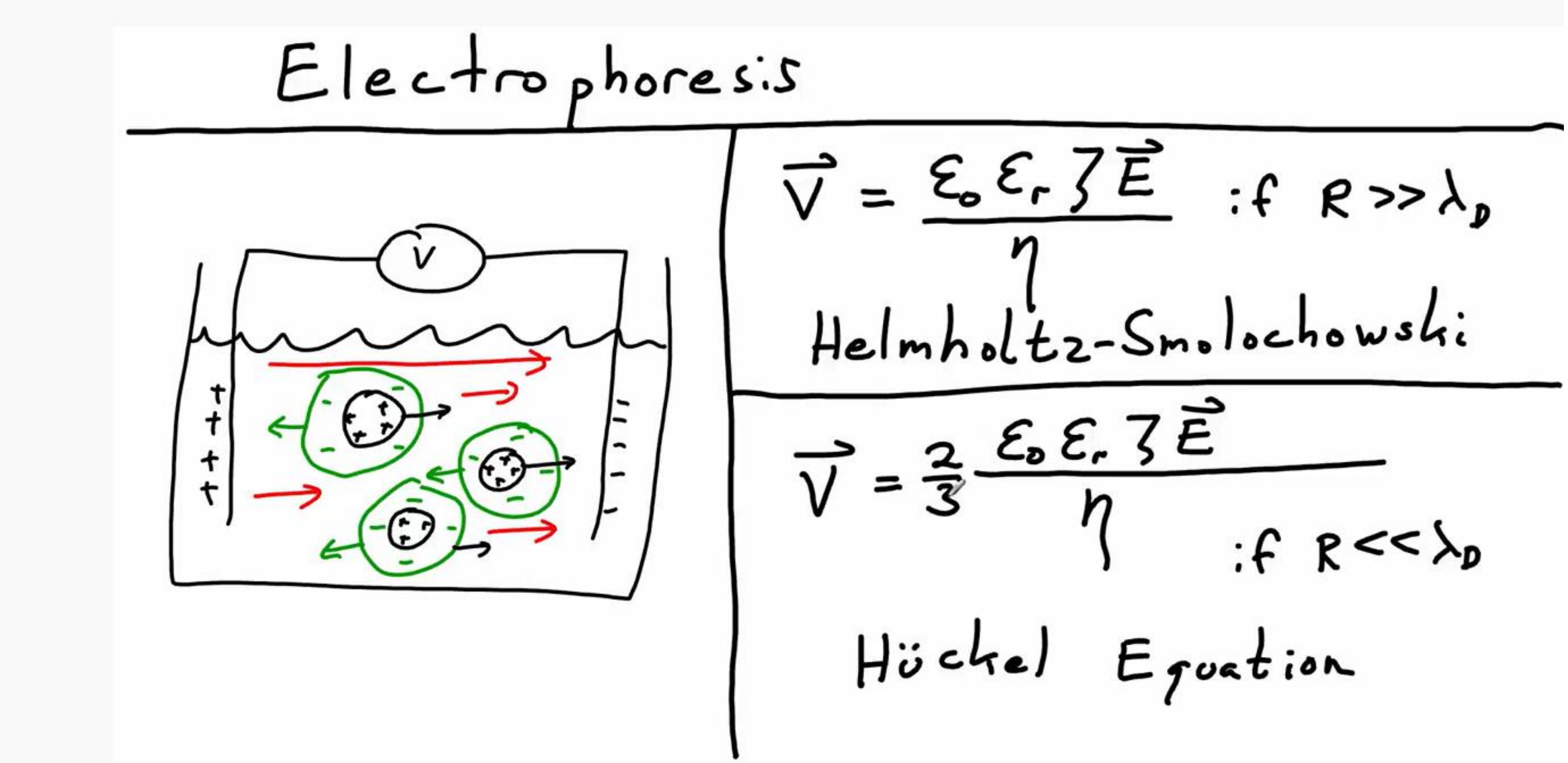
ζ = zeta potential, E = electric field

η = medium viscosity

μ_E = electrophoretic mobility

ZETA POTENTIAL MEASUREMENT

Electrophoresis



ZETA POTENTIAL MEASUREMENT

Electrophoresis

Henry Formula (1931)

expanded for arbitrary κa, assumed E-field does not alter surface charge

$$\vec{v} = \frac{2\varepsilon_r \varepsilon_0 \zeta}{3\eta} f_1(\kappa a)\vec{\mathrm{E}}$$

$$\mu_{\mathrm{E}} = \frac{2\varepsilon_r \varepsilon_0 \zeta}{3\eta} f_1(\kappa a)$$

v = velocity

ε_r = media dielectric constant

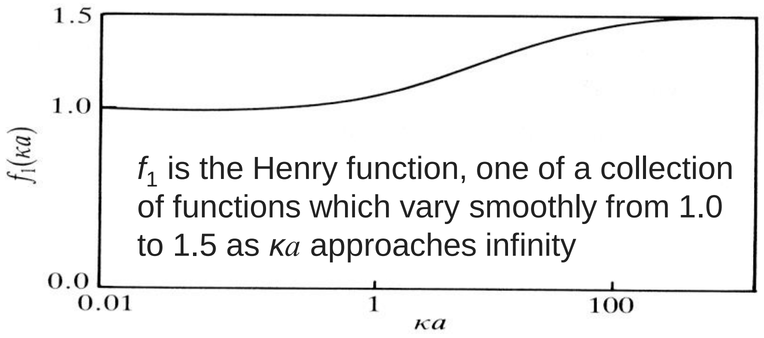
ε_0 = permittivity of free space

ζ = zeta potential

η = medium viscosity

E = electric field

μ_E = electrophoretic mobility

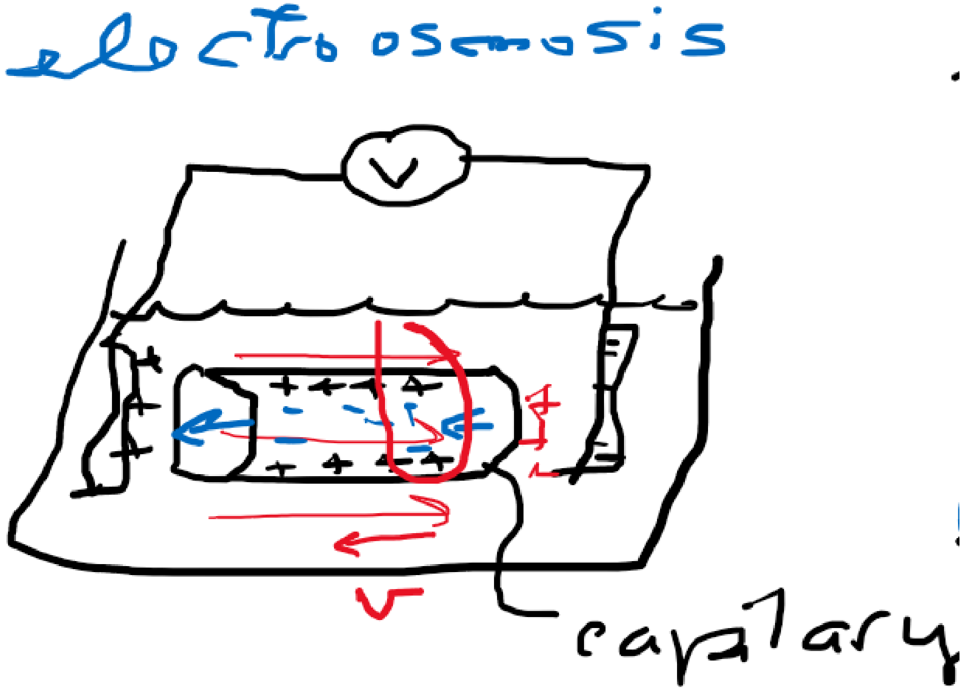


f_1 is the Henry function, one of a collection of functions which vary smoothly from 1.0 to 1.5 as κa approaches infinity

Hunter, Foundations of Colloid Science, p. 560

ZETA POTENTIAL MEASUREMENT

Electro-osmosis



$$\vec{v} = -\frac{\varepsilon_r \varepsilon_0 \zeta}{\eta} \vec{\mathrm{E}}$$

v = velocity,

ε_r = media dielectric constant

ε_0 = permittivity of free space

ζ = zeta potential, E = electric field

η = medium viscosity

ZETA POTENTIAL MEASUREMENT

Streaming Potential

$$\frac{\exp(Ze\zeta/2kT)}{\kappa a} \ll 1$$

κ = Debye parameter
a = particle radius
ζ = zeta potential

$$\Delta V = \frac{\varepsilon_r \varepsilon_0 \zeta}{\eta K_E} \Delta p$$

ΔV = Potential over capillary (V)
ε_r = media dielectric constant
ε_0 = permittivity of free space (F/m)
η = medium viscosity (Pa·s)
K_E = solution conductivity (S/m)
Δp = pressure drop across capillary (Pa)

ZETA POTENTIAL MEASUREMENT

Sedimentation Potential

$$V = \frac{l\varepsilon_r\varepsilon_0\left(\rho-\rho_0\right)g\zeta}{\eta K_E}$$

l = height of the tube

ζ = zeta potential

V = Sedimentation Potential (V)

ε_r = media dielectric constant

ε_0 = permittivity of free space (F/m)

η = medium viscosity (Pa·s)

K_E= solution conductivity (S/m)

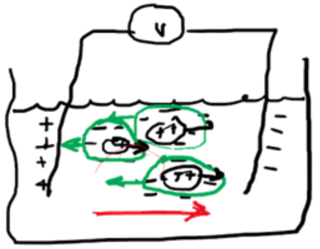
ρ = density of colloidal particles

ρ_0 = density of solution

ELECTROKINETIC PHENOMENA

Universidade do Minho

electrophoresis



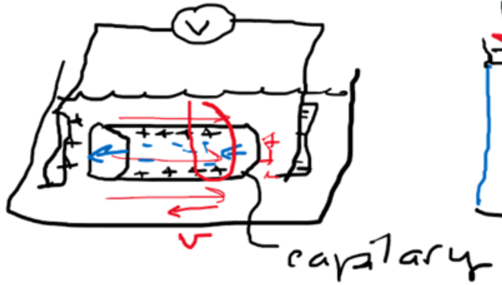
$$v = \frac{\varepsilon_r \varepsilon_0 \zeta}{\eta} E$$

$$\mu_E = \frac{\varepsilon_r \varepsilon_0 \zeta}{\eta}$$

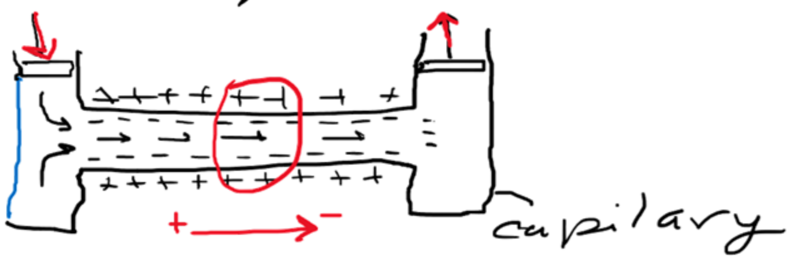
v = velocity,

ε_r = media dielectric constant
ε_0 = permittivity of free space
ζ = zeta potential, E = electric field
η = medium viscosity
μ_E = electrophoretic mobility

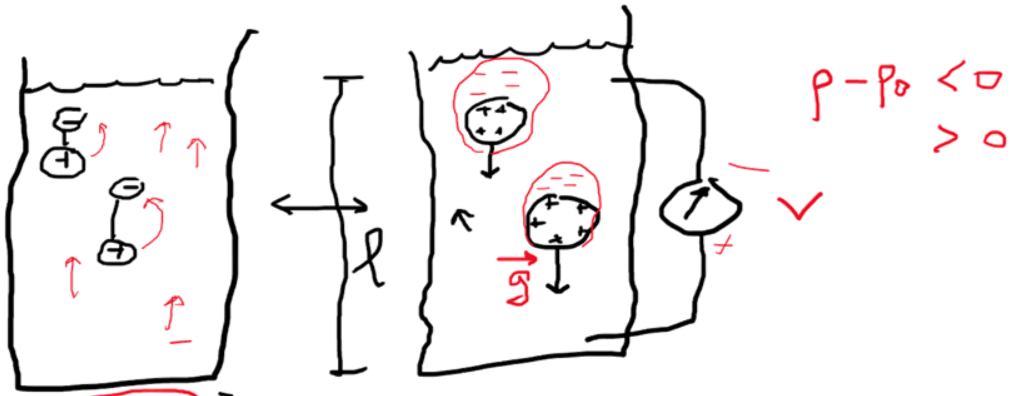
electroosmosis



streaming potential



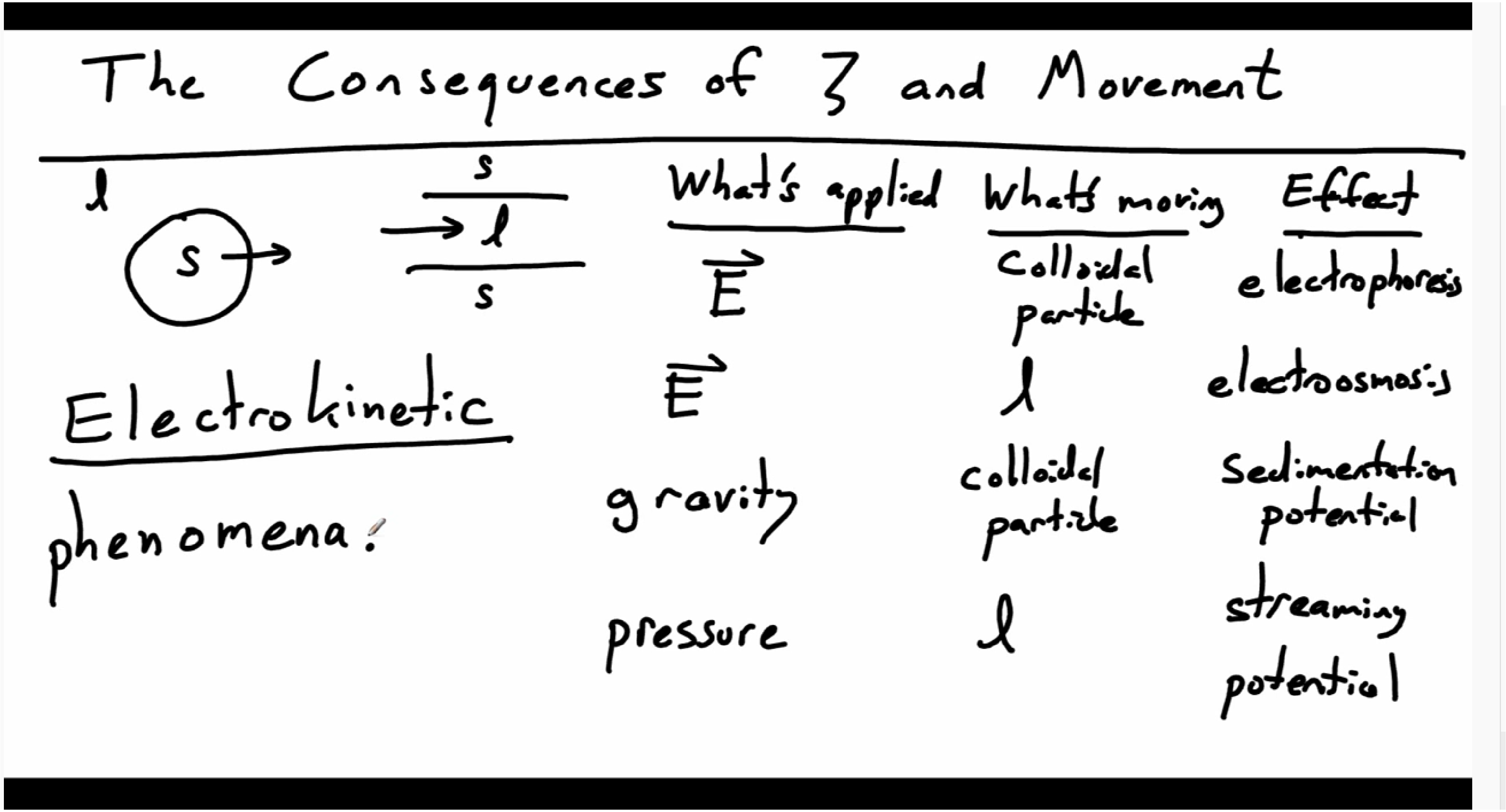
$$\Delta E = \frac{\varepsilon_r \varepsilon_0 \zeta}{\eta K_E} \Delta p$$



$$v = \frac{l \varepsilon_r \varepsilon_0 (\rho - \rho_0) g \zeta}{\eta \cdot K_E}$$

sedimentation potential

ELECTROKINETIC PHENOMENA



Combined Effects of van der Waals and Electrostatic Forces

DLVO Theory

DLVO – Derjaguin, Landau, Verwey and Overbeek

Based on the sum of van der Waals attractive potential and a screened electrostatic repulsion potential arising between the "double layer potential" screened by ions in solution. The total interaction energy U of the system is:

$$U(x) = -\frac{H}{12\pi x^2} + \frac{64nkTR\gamma^2}{\kappa}\exp(-\kappa x)$$

Flat surfaces

Van der Waals (Attractive force)

Electrostatics (Repulsive force)

DLVO Theory

Van der Waals interaction

EDL interaction

$$U(x) = -\frac{HR}{12x} + \frac{64\pi nkTR\gamma^2}{\kappa^2}\exp(-\kappa x)$$

Identical spherical particles

$$\gamma = \tanh\left(\frac{ze\psi}{4kT}\right)$$

H **= Hamakar's constant**

R **= Radius of particle**

x **= Distance of Separation**

k **= Boltzmann's constant**

T **= Temperature**

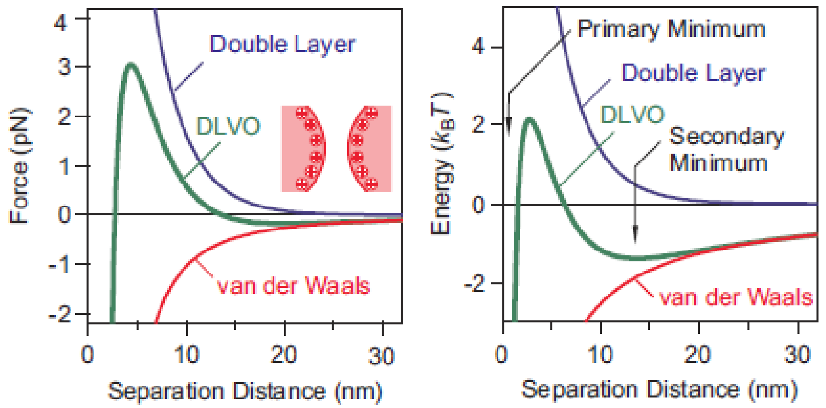
n **= bulk ion concentration**

κ= Debye parameter

z **= valency of ion**

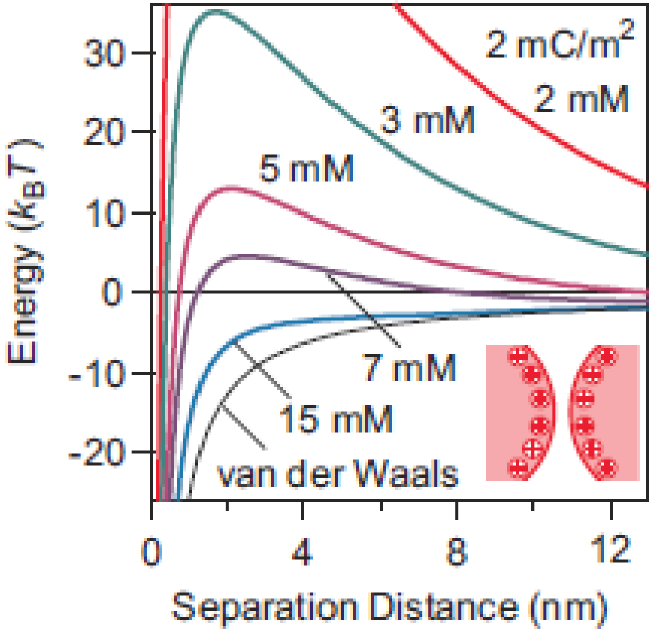
e **= Charge of electron**

Ψ **= Surface potential**

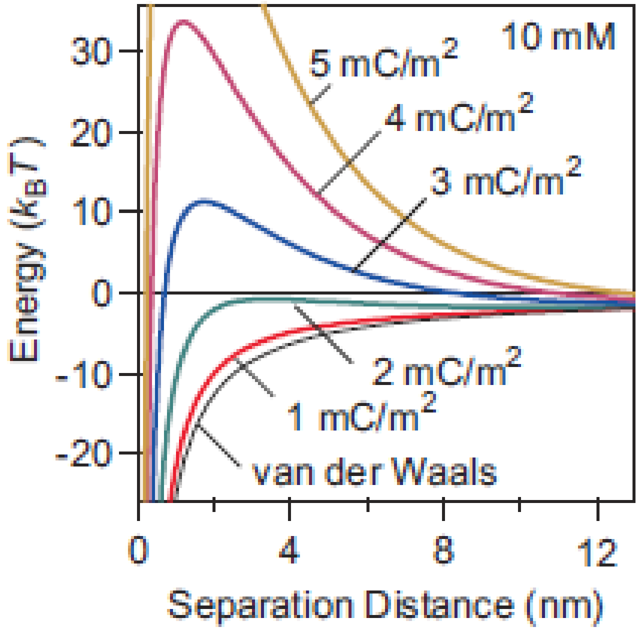


DLVO in practice

dependence on the salt concentration

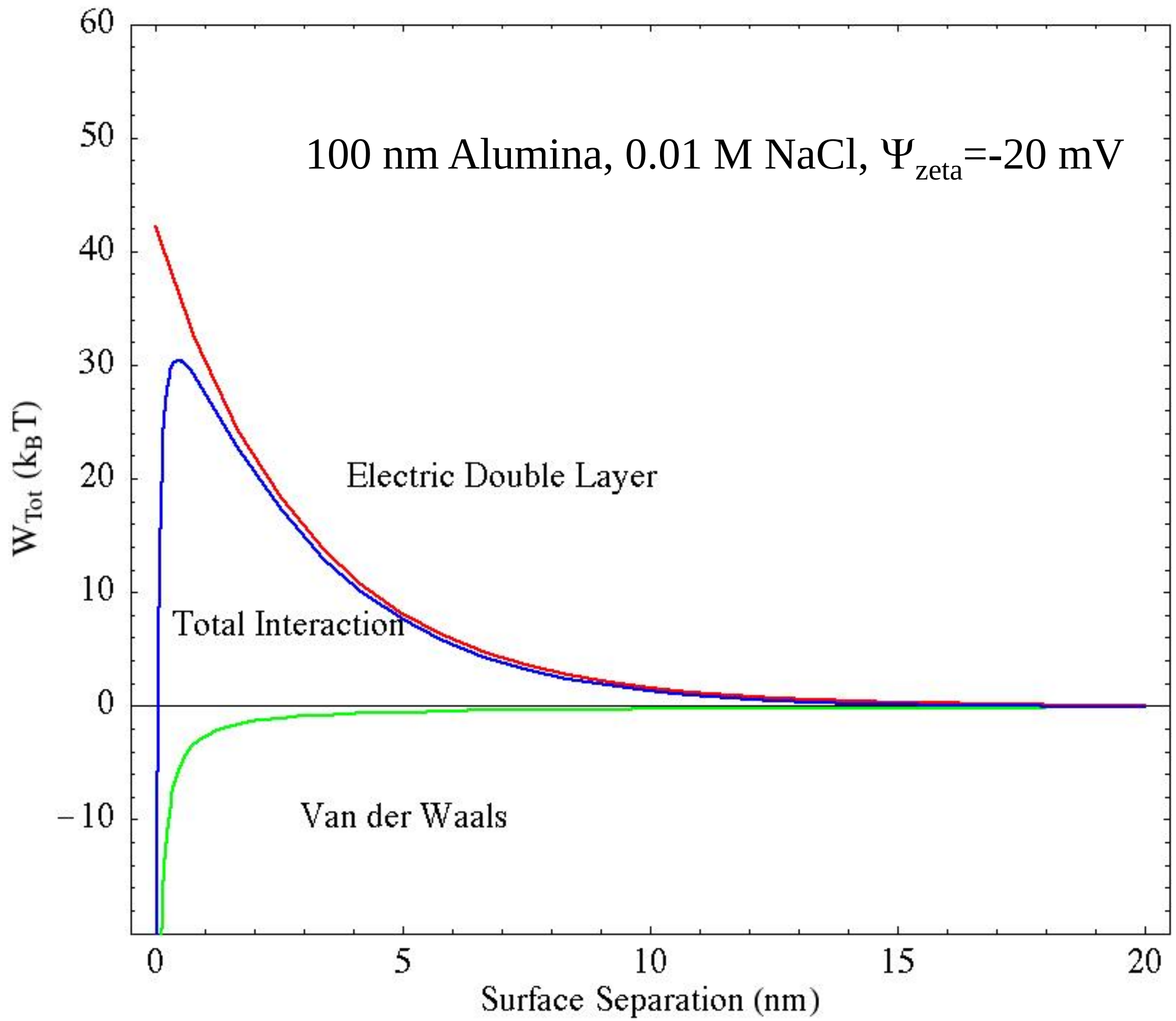


dependence on surface charge density



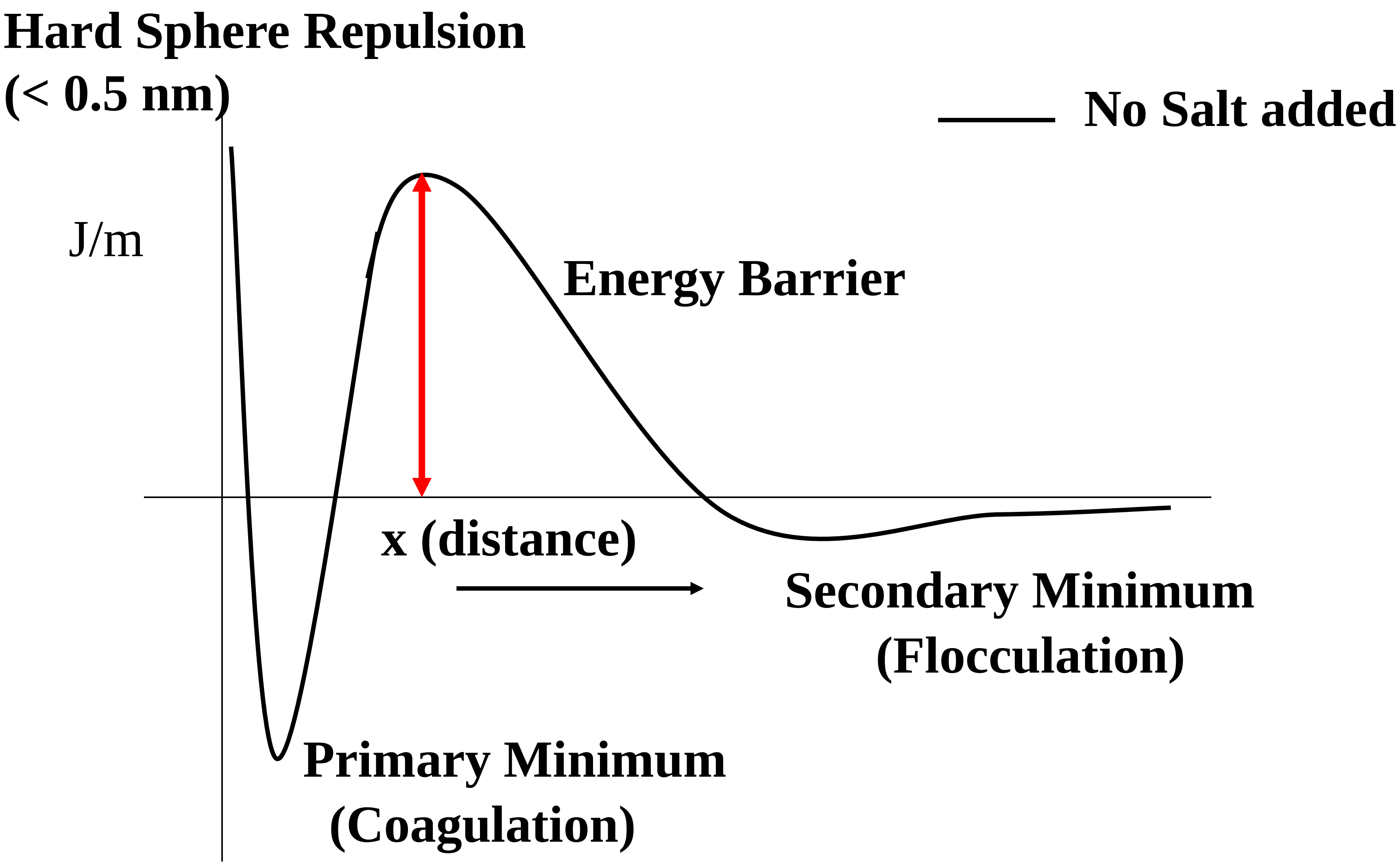
- At low salt levels or high surface charge densities, the particles are repulsive, as the interaction is dominated by the double layer contribution
- At high salt levels or small surface charge densities, the interaction is dominated by the attractive van der Waals force.
- At intermediate values, the energy profile passes through a maximum, which occurs at a separation distance that is comparable to the Debye length κ^{-1}

DLVO Theory



For short distances of separation between particles

DLVO Theory


Hard Sphere Repulsion
(< 0.5 nm)
No Salt added
J/m
Energy Barrier
x (distance)
Secondary Minimum
(Flocculation)
Primary Minimum
(Coagulation)

Discussion: Flocculation vs. Coagulation

The DLVO theory defines formally (and distinctly), the often inter-used terms flocculation and coagulation

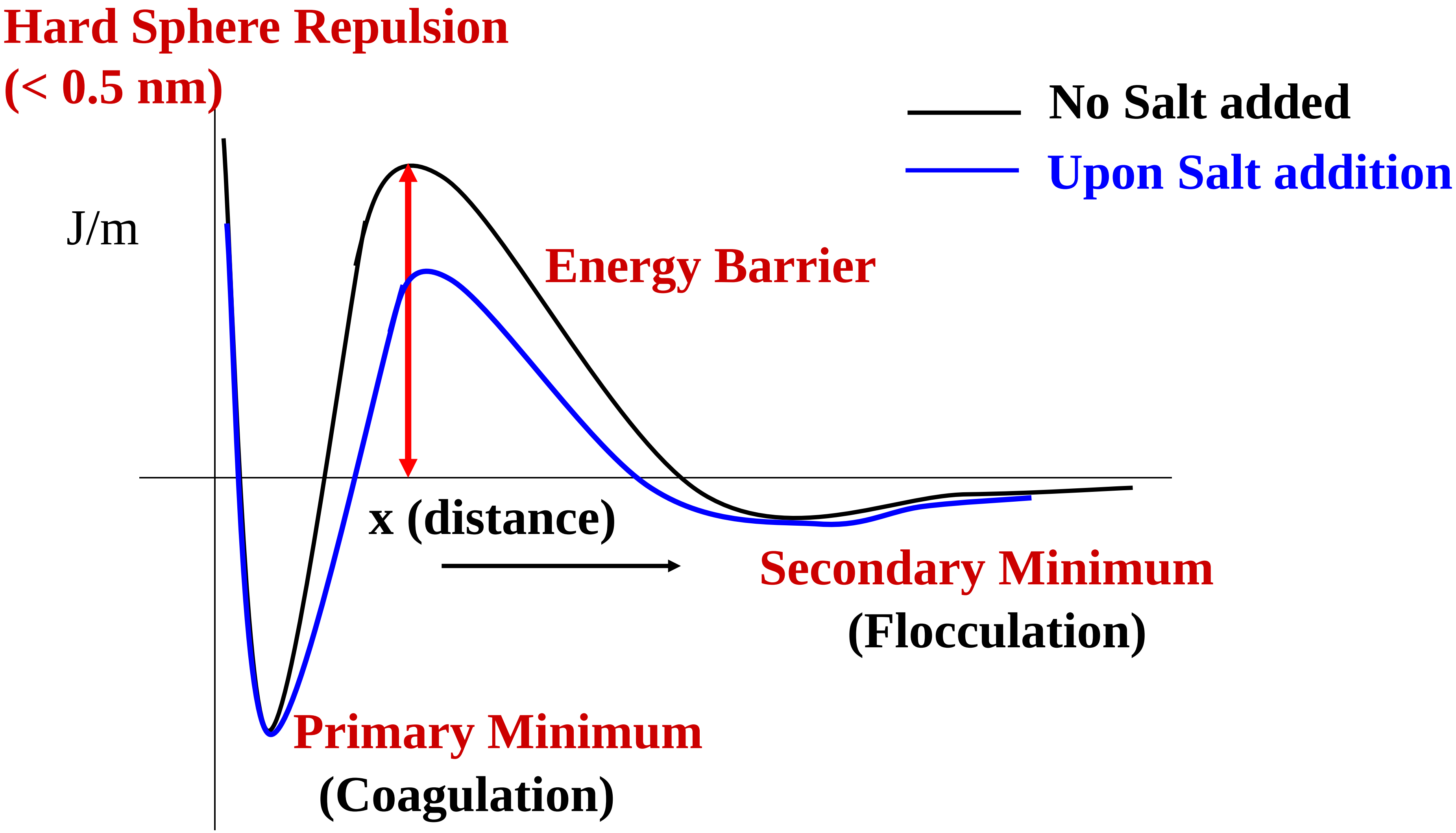
<u>Flocculation:</u>

- Corresponds to the secondary energy minimum at large distances of separation
- The energy minimum is shallow (weak attractions, 1-2 kT units)
- Attraction forces may be overcome by simple shaking

<u>Coagulation:</u>

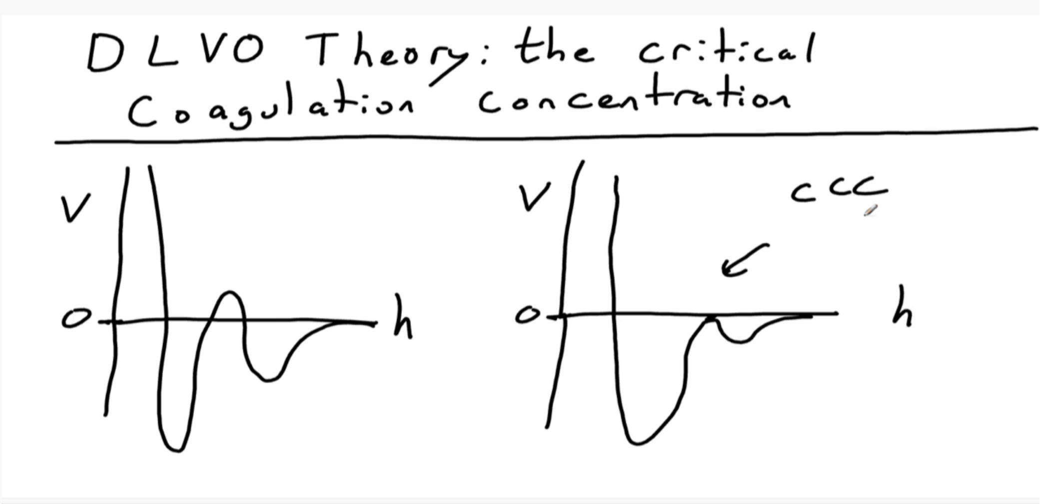
- Corresponds to the primary energy minimum at short distances of separation upon overcoming the energy barrier
- The energy minimum is deep (strong attractions)
- Once coagulated, particle separation is almost impossible

Effect of Salt



Addition of salt reduces the energy barrier of repulsion.
How?

Discussion on the Effect of Salt



- The salt reduces the EDL thickness by charge screening
- Reduces the energy barrier (may induce coagulation)
- Also increases the distance at which secondary minimum occurs (aids flocculation)
- Since increased salt concentration decreases κ^{-1} (or decreases electrostatics), at the Critical Salt Concentration $U(x) = 0$